MOG JEE Main Full Test - 2024 Revised

Total Time: 180 minutes | Total Marks: 300 | Total Questions: 75

GENERAL INSTRUCTIONS

The Question Paper consists of three parts (Physics, Chemistry and Mathematics) with a total of **75** questions.
Each subject is divided into 2 sections:
Section 1: Single Correct Type Questions (4 options, only one correct)
Section 2: Numerical Value Type Questions (answer to be entered as numerical value)
Marking Scheme:
Single Correct: +4 for correct, -1 for wrong, 0 if unattempted.
Numerical Value: +4 for correct, -1 for wrong.

By: C I P H Ξ R

Q.1

For the equation $F \propto A^a v^b d^c$, where F is the force, A is the area, v is the velocity and d is the density, the values of a, b and c are respectively

(A) $1, 2, 1$

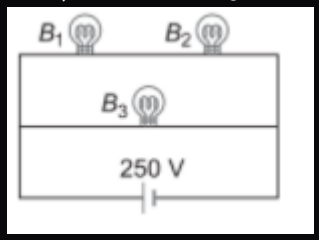
(B) $2, 1, 1$

(C) $1, 1, 2$

(D) $0, 1, 1$

Q.2

A $100\ \text{W}$ bulb B_1 and two $60\ \text{W}$ bulb B_2 and B_3 are connected to a $250\ \text{V}$ source as shown in the figure. Now W_1, W_2 and W_3 are the output powers of the bulbs B_1, B_2 and B_3 respectively. Then



(A) $W_1 > W_2 = W_3$

(B) $W_1 > W_2 > W_3$

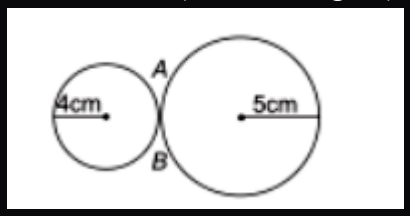
(C) $W_1 < W_2 = W_3$

(D) $W_1 < W_2 < W_3$

Q.3

Two soap bubbles of radii r_1 and r_2 equal to $4\ \text{cm}$ and $5\ \text{cm}$ respectively are touching each other over a common

surface AB (shown in figure). Its radius will be



(A) $4\ \text{cm}$

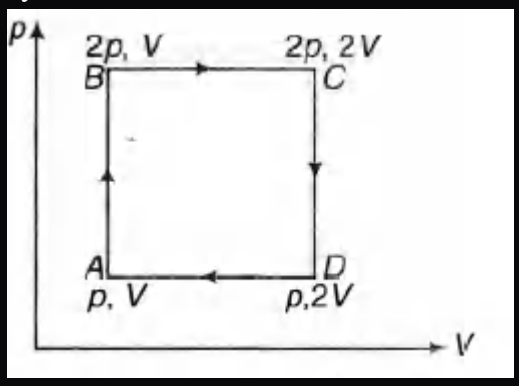
(B) $4.5\ \text{cm}$

(C) $5\ \text{cm}$

(D) $20\ \text{cm}$

Q.4

An ideal monoatomic gas is taken around the cycle $ABCD$ as shown in p versus V diagram. Work done during the cycle is



(A) pV

(B) $0.5\ \text{pV}$

(C) $2\ \text{pV}$

(D) $3\ \text{pV}$

Q.5

Mass of Bromine, Selenium and Krypton are given as :-

$\text{M}\left({}^{80}{}_{\text{Br}}\right) = 79.918528\text{u}, \text{M}\left({}^{80}\text{Kr}\right) = 79.916376\text{u},$

$\text{M}\left({}^{80}\text{Se}\right) = 79.916521\text{u}$

Which of the following β decays are possible?

(i) ${}^{80}_{35}\text{Br} \rightarrow {}^{80}_{36}\text{Kr} + {}^{0}_{-1}\text{e} + \vec{v}$

(ii) ${}^{80}_{35}\text{Br} \rightarrow {}^{80}_{34}\text{Se} + {}^{0}_{+1}\text{e} + v$

(iii) ${}^{80}_{35}\text{Br} + {}^{0}_{-1}\text{e} \rightarrow {}^{80}_{34}\text{Se} + v$

(A) all 3

(B) (i) & (iii) only

(C) (ii) & (iii) only

(D) (i) & (ii) only

Q.6

Assertion: No work is done in moving an electric dipole transnationally in uniform electric field.
Reason: Net force on electric dipole in uniform electric field is zero.

(A) If both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(B) If Assertion is true but Reason is false.

(C) If Assertion is false but Reason is true.

(D) If both Assertion and Reason are false.

Q.7

Consider the following statements about a solenoid.
1. The magnetic field strength in a solenoid depends upon the number of turns per unit length in the solenoid.
2. The magnetic field strength in a solenoid depends upon the current flowing in the wire of the solenoid.
3. The magnetic field strength in a solenoid depends upon the diameter of the solenoid.
Which of the statements given above are correct?

(A) 1, 2 and 3

(B) 1 and 3

(C) 2 and 3

(D) 1 and 2

Q.8

Vernier caliper has main scale division of 1 mm. It has 20 equal divisions on the vernier scale which match with 16 main scale division. For this vernier caliper, the least count is:

(A) $0.02\ \text{mm}$

(B) $0.2\ \text{mm}$

(C) $0.05\ \text{mm}$

(D) $0.1\ \text{mm}$

Q.9

A parallel plate capacitor is charged and then disconnected from the source of potential difference. If the plates of the condenser are moved farther apart by the use of insulated handle, which of the following is true?

(A) The charge on the capacitor increases

(B) The charge on the capacitor decreases

(C) The capacitance of the capacitor increases

(D) The potential difference across the plates increases

Q.10

A sample of hydrogen like atoms produces an emission spectrum consisting of 10 wavelength arising out of all transitions possible. During this process the maximum angular impulse on an electron going from higher energy level to a lower energy level is :-

(A) $\frac{\mathrm{h}}{2\pi}$

(B) $\frac{\mathrm{h}}{\pi}$

(C) $\frac{2\mathrm{h}}{\pi}$

(D) $\frac{3\mathrm{h}}{2\pi}$

Q.11

Two planets have masses M and $16\mathrm{M}$ and their radii are a and $2\mathrm{a}$, respectively. The separation between the centres of the planets is $10\mathrm{a}$. A body of mass m is fired from the surface of the larger planet towards the smaller planet along the line joining their centres. For the body to be able to reach at the surface of smaller planet, the minimum firing speed needed is :

(A) $2\sqrt{\frac{\mathrm{GM}}{\mathrm{a}}}$

(B) $4\sqrt{\frac{\mathrm{GM}}{\mathrm{a}}}$

(C) $\sqrt{\frac{\mathrm{GM}^2}{\mathrm{ma}}}$

(D) $\frac{3}{2}\sqrt{\frac{5\mathrm{GM}}{a}}$

Q.12

A particle is projected from ground at some angle with horizontal (Assuming point of projection to be origin and the horizontal and vertical directions to be the x and $\mathrm{y}-$ axis) the particle passes through the points $(3\ \mathrm{m},\ 4\ \mathrm{m})$ and $(4\ \mathrm{m},\ 3\ \mathrm{m})$ during its motion then the range of the particle would be: $\left(g=10\ \mathrm{m\ s^{-2}}\right)$

(A) $36\ \mathrm{m}$

(B) $\frac{37}{7}\ \mathrm{m}$

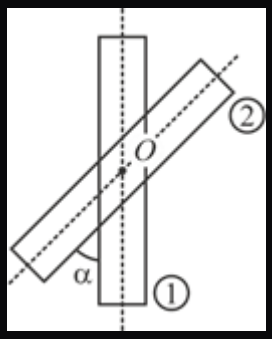
(C) $4\ \mathrm{m}$

(D) $16\ \mathrm{m}$

Q.13

Two identical rods of mass M and length l are placed in a horizontal plane with a common centre O as shown in the figure. Moment of inertia of the system about an axis passing through O and perpendicular to the horizontal plane

will be



(A) $\frac{Ml^2}{6}$

(B) $\frac{Ml^2}{3}$

(C) $\frac{Ml^2}{6}(1+\alpha)$

(D) $\frac{Ml^2}{3}(\alpha+3)$

Q.14

Match **List - I** (Electromagnetic wave type) with **List - II** (Its association/application) and select the correct option from the choices given below the lists :

	List - I		List - II
(a)	Infrared waves	(i)	To treat muscular strain
(b)	Radio waves	(ii)	For broadcasting
(c)	X - rays	(iii)	To detect fracture of bones
(d)	Ultraviolet rays	(iv)	Absorbed by the ozone layer of the atmosphere

(A) (a) (b) (c) (d)
(iv) (iii) (ii) (i)

(B) (a) (b) (c) (d)
(i) (ii) (iv) (iii)

(C) (a) (b) (c) (d)
(iii) (ii) (i) (iv)

(D) (a) (b) (c) (d)
(i) (ii) (iii) (iv)

Q.15

In the double slit experiment, 5 th dark fringe is formed opposite to one of the slits. The wavelength of light is

(A) $\frac{d^2}{6D}$

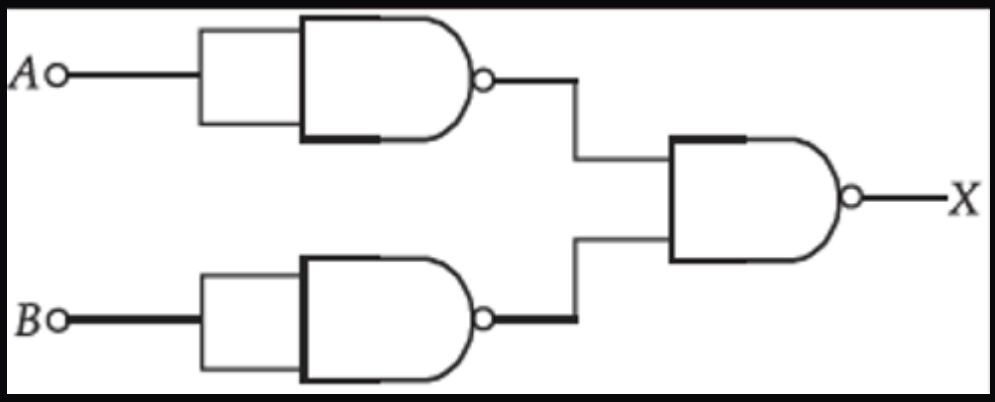
(B) $\frac{d^2}{5D}$

(C) $\frac{d^2}{15D}$

(D) $\frac{d^2}{9D}$

Q.16

The combination of gates shown below yields



(A) OR gate

(B) NOT gate

(C) XOR gate

(D) NAND gate

Q.17

A steel wire of length l has a magnetic moment M. It is then bent into a semicircular arc. The new magnetic moment is

(A) $\frac{M}{\pi}$

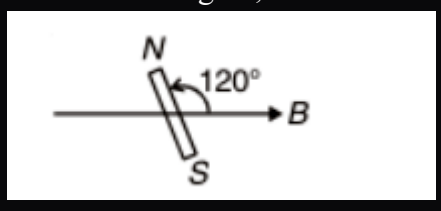
(B) $\frac{2M}{\pi}$

(C) $\frac{3M}{\pi}$

(D) $\frac{4M}{\pi}$

Q.18

A magnet of magnetic dipole moment M is released in a uniform magnetic field of induction B. from the position shown in the figure, then the maximum kinetic energy during motion of needle is



(A) $\frac{1}{2}MB$

(B) MB

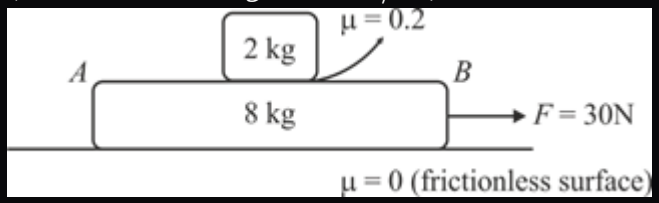
(C) $\frac{3}{2}MB$

(D) $2MB$

Q.19

Two blocks are placed as shown in the figure. The upper block is placed at $2\ \mathrm{m}$ from the end A. Find time taken by

$2\ \text{kg}$ box to fell from A.
(Take the value of g as 10m/s^2)



(A) $\sqrt{3}s$

(B) $2\ \text{s}$

(C) $3\ \text{s}$

(D) $1.2\ \text{s}$

Q.20

A cylindrical metallic rod in thermal contact with two reservoirs of heat at its two ends conducts an amount of heat Q in time t. The metallic rod is melted and the material is formed into a rod of half the radius of the original rod. What is the amount of heat conducted by the new rod, when placed in thermal contact with the two reservoirs in time t?

(A) $\frac{Q}{4}$

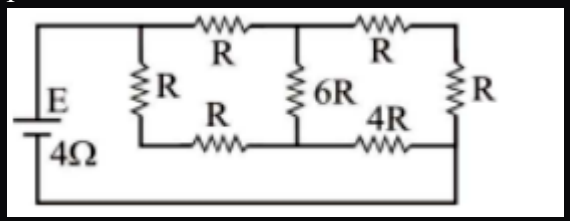
(B) $\frac{Q}{16}$

(C) $2Q$

(D) $\frac{Q}{2}$

Q.21

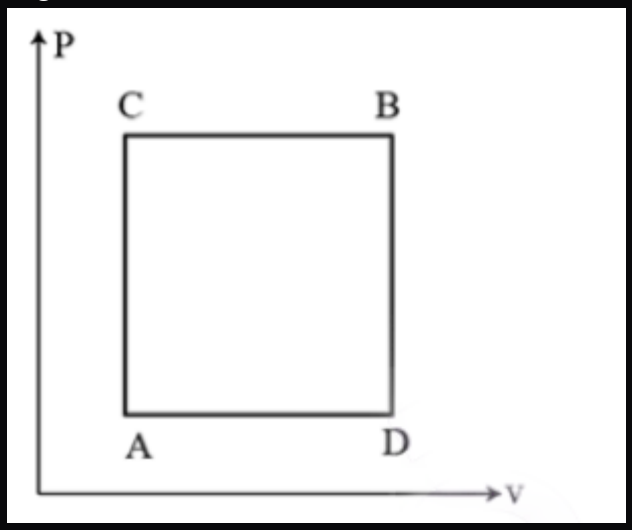
A battery of internal resistance $4\ \Omega$ is connected to the network of resistances as shown. In order that the maximum power can be delivered to the network, the value of R in Ω should be



(A) Submit

Q.22

A gas can be taken from A to B via two different processes ACB and ADB.



When path ACB is used 60 J of heat flows into the system and 30 J of work is done by the system. If path ADB is used work done by the system is 10 J. The heat flow (in joule) into the system in path ADB is :

(A) Submit

Q.23

A point source S is placed at a distance of 15 cm from a converging lens of focal length 10 cm on its principal axis. The diverging mirror of focal length 12 cm is placed at a certain distance from lens such that a real image is formed on the source itself. Find the minimum distance.

(A) Submit

Q.24

A solid sphere of radius R has a charge Q distributed in its volume with a charge density $\rho = kr^a$, where k and a are constants and r is the distance from its centre. If the electric field at $r = \frac{R}{2}$ is $\frac{1}{8}$ times that at $r = R$, find the value of a.

(A) Submit

Q.25

A mass 0.9 kg, attached to a horizontal spring, executes SHM with an amplitude A_1. When this mass passes through its mean position, then a smaller mass of 124 g is placed over it and both masses move together with amplitude A_2. If the ratio $\frac{A_1}{A_2}$ is $\frac{\alpha}{\alpha-1}$, then the value of α will be ____.

(A) Submit

Q.26

Which of the following ions has the highest magnetic moment

(A) Ti^{3+}

(B) Sc^{3+}

(C) Mn^{2+}

(D) Zn^{2+}

Q.27

A $KMnO_4$ solution can be standardised by titration against $As_2O_3(s)$. A 0.1156 g sample of As_2O_3 requires 27.08 mL of the $KMnO_4$ (aq) for its titration. What is the molarity of the $KMnO_4(aq)$ [As = 75].

(A) 0.0172 M

(B) 1.0172 M

(C) 0.172 M

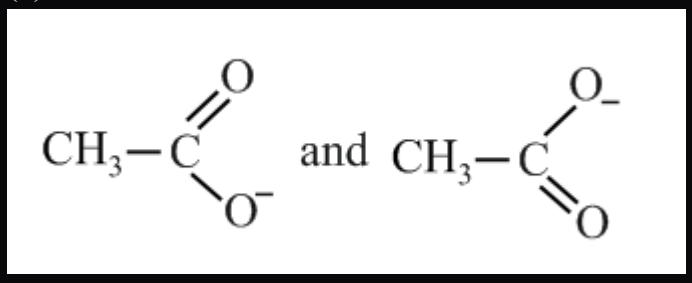
(D) 0.9172 M

Q.28

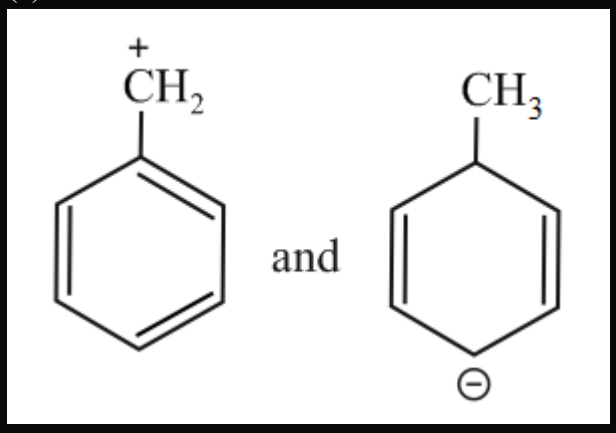
Which of the following chemical species are not related through resonance?

(a) $CH_3COCH_2COOC_2H_5$ and $CH_3-\overset{OH}{\overset{|}{C}}=CHCOOCH_2H_5$

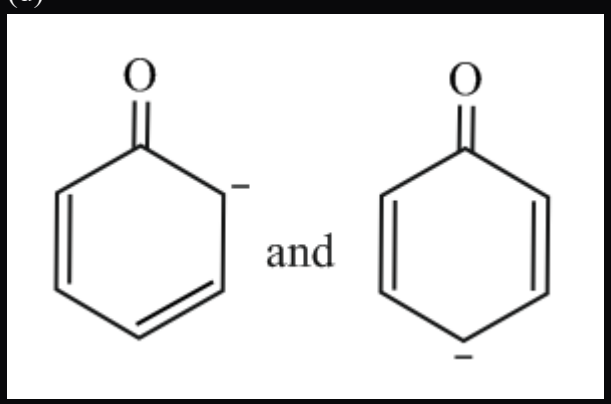
(b)



(c)



(d)



(A) Both (a) and (d)

(B) Both (a) and (c)

(C) Both (a) and (b)

(D) Both (b) and (d)

Q.29

Assertion: PbO_2 is an oxidising agent and reduced to PbO.
Reason: Stability of $Pb(II) > Pb(IV)$ on account of inert pair effect.

(A) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(B) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

(C) Assertion is true but Reason is false.

(D) Assertion is false but Reason is true.

Q.30

Match list I (reagents for reaction of nitrobenzene) with list II (products formed) and select the correct answer using the codes given below.

	List I		List II
(p)	Sn and HCl	1.	Hydrazobenzene
(q)	Zn and NH4Cl	2.	Azoxybenzene
(r)	Methanolic NaOMe	3.	Phenyl hydroxylamine
(s)	Zn and KOH	4.	Aniline

(A) (p)-4, (q)-3, (r)-1, (s)-2

(B) (p)-4, (q)-3, (r)-2, (s)-1

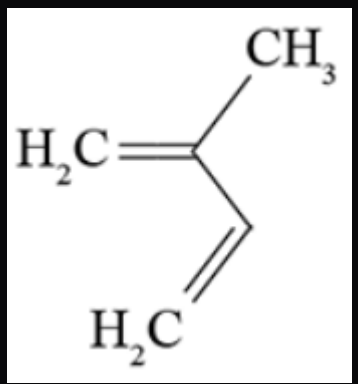
(C) (p)-3, (q)-4, (r)-1, (s)-2

(D) (p)-3, (q)-4, (r)-2, (s)-1

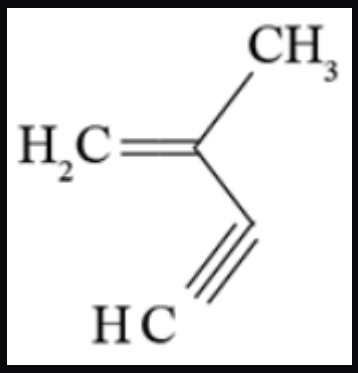
Q.31

An organic compound A (C_5H_8) on hydrogenation gives compound B (C_5H_{12}) compound A on ozonolysis give methanal and 2 -ketopropanal. The compound A is

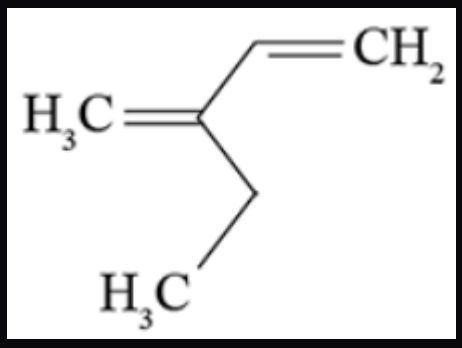
(A)



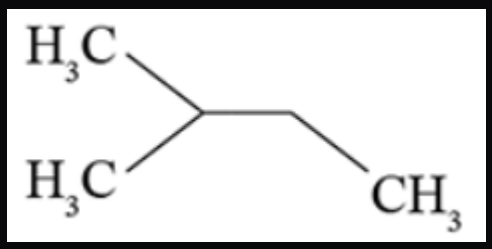
(B)



(C)



(D)



Q.32

Which is not paramagnetic?

(A) O_2

(B) O_2^+

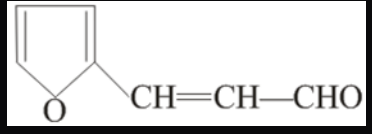
(C) O_2^-

(D) O_2^{2-}

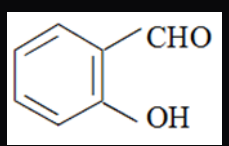
Q.33

An aromatic compound 'P' having molecular formula $C_7H_6O_2$ on treating with aqueous ammonia and heating forms compound 'Q'. The compound 'Q' on reaction with molecular bromine and potassium hydroxide provides compound 'R' having molecular formula C_6H_7N . The structure of compound 'P' is

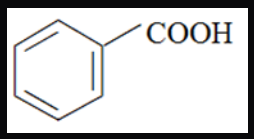
(A)



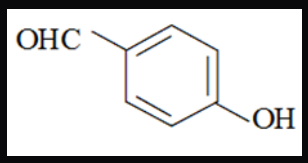
(B)



(C)



(D)



Q.34

Al^{3+} has low ionic radius than Mg^{2+} because

(A) Al^{3+} has high nuclear charge than Mg^{2+}

(B) Mg atom has less no. of neutrons than Al atom

(C) Mg and Al differ in electronegativity values

(D) Al atom has low msub value than Mg atom

Q.35

What will be the volume of O_2 Liberated at NTP by passing 5 A current For 193 Sec. through acidified water.

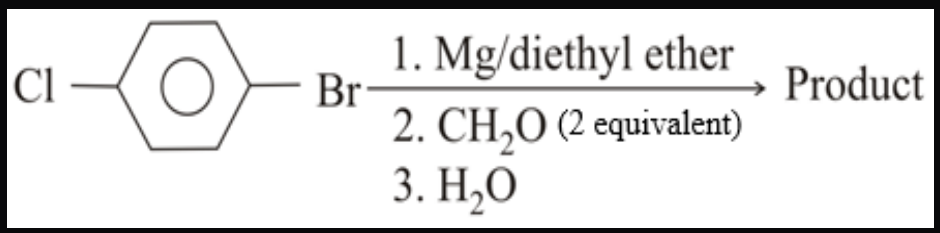
(A) 56 mL

(B) 112 mL

(C) 158 mL

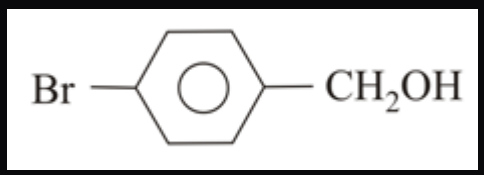
(D) 965 mL

Q.36

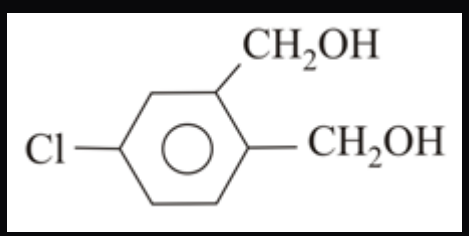


The product in the above reaction is

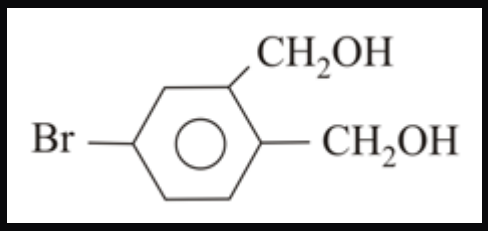
(A)



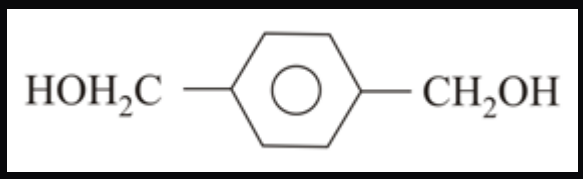
(B)



(C)



(D)



Q.37

If a transition-metal compound absorbs violet radiation in the visible region, its colour would be:

(A) Red

(B) Yellow

(C) Orange

(D) Blue

Q.38

Directions : Each of these questions contains an Assertion followed by reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.
Assertion: o-Hydroxy benzene sulphonic acid on heating to $373\ \mathrm{K}$ changes to p-Hydroxybenzene sulphonic acid.
Reason: Sulphonation of phenol is a reversible process.

(A) If both assertion and reason are true and reason is the correct explanation of assertion.

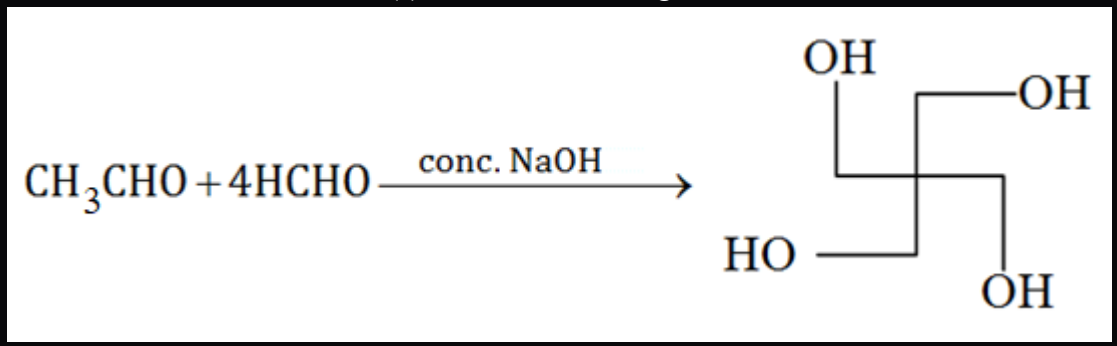
(B) If both assertion and reason are true but reason is not the correct explanation of assertion.

(C) If Assertion is true but reason is false.

(D) If both assertion and reason are false.

Q.39

The number of aldol reaction (s) that occurs in the given transformation is



(A) 5

(B) 3

(C) 2

(D) 4

Q.40

Among the following molecules/ions, what is the correct order of increasing s-character (in percentage) in the hybrid orbitals?

(I) CO_3^{2-} (II) XeF_4 (III) I_3^- (IV) NCl_3 (V) $BeCl_2$

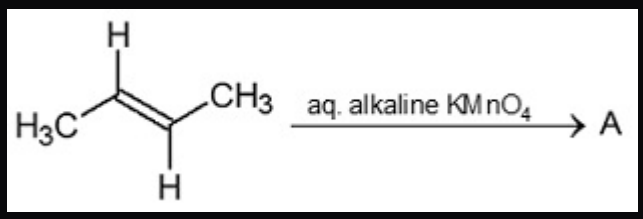
(A) $\mathrm{II} < \mathrm{III} < \mathrm{IV} < \mathrm{I} < \mathrm{V}$

(B) $\mathrm{II} < \mathrm{IV} < \mathrm{III} < \mathrm{V} < \mathrm{I}$

(C) $\mathrm{III} < \mathrm{II} < \mathrm{I} < \mathrm{V} < \mathrm{IV}$

(D) $\mathrm{II} < \mathrm{IV} < \mathrm{III} < \mathrm{I} < \mathrm{V}$

Q.41



Which one of the following is true about this reaction?

(A) A is meso-2, 3-butanediol formed by syn addition

(B) A is meso-2, 3-butanediol formed by anti-addition

(C) A is a racemic mixture of d and l-2, 3-butanediol formed by anti-addition

(D) A is a racemic mixture of d and l-2, 3-butanediol formed by syn addition

Q.42

The sequence of an mRNA molecule produced from a DNA template strand with the composition $5' - \mathrm{AGCTACACT} - 3'$ is

(A) 5 '-AGUGUAGCU-3'

(B) 5'-UCGAUGUGA-3'

(C) $5'$-AGTGTAGCT-3'

(D) $5'$-TCGATGTGA-3'

Q.43

The composition of wustite is $\mathrm{Fe}_{0.95}\mathrm{O}_1$. What is the percentage of iron as Fe^{3+} ions?

(A) 10.5

(B) 25

(C) 35

(D) 45

Q.44

Maximum amount of a solid solute that can be dissolved in a given amount of a liquid solvent does not depend upon

(A) pressure

(B) temperature

(C) nature of solute

(D) nature of solvent

Q.45

In an atom, two electrons revolve around the nucleus in circular orbits of radii r and $4r$. The ratio of the time taken by them $(\mathrm{T}_2 : \mathrm{T}_1)$ to complete one revolution is

(A) $1 : 4$

(B) $4 : 1$

(C) $1 : 8$

(D) $8 : 1$

Q.46

In general, dissociation constant of a weak monoprotic acid is numerically equal to the dissociation constant of its conjugate base. The pH of 0.1 M solution of this acid is

(A) Submit

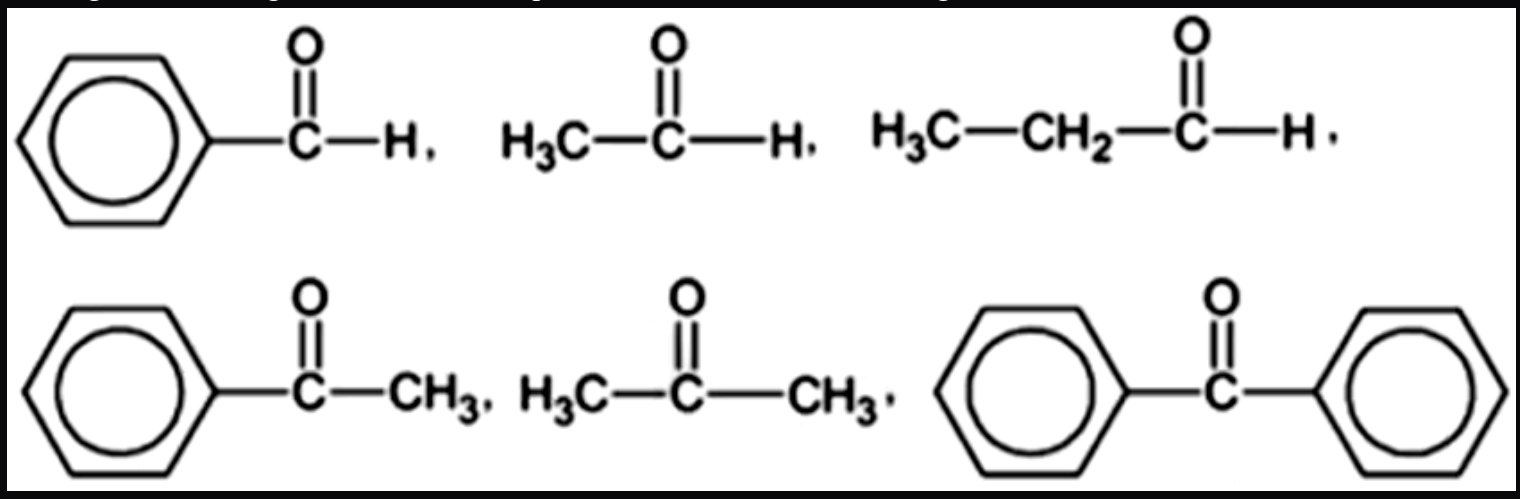
Q.47

Total number of optically inactive stereoisomers of succinic acid are

(A) Submit

Q.48

Among the following the number of compounds which react with Fehling solution is:



(A) Submit

Q.49

The polymerisation of polypropene to linear polypene is represented by the reaction $\mathrm{n}\left(\begin{array}{l}\mathrm{CH_3} \\ \mathrm{CH=CH_2}\end{array}\right) \longrightarrow \left(\begin{array}{l}\mathrm{CH_3} \\ \mathrm{CH-CH_2}\end{array}\right)_{\mathrm{n}}$ Where n has large integral value, the average enthalpies of bond dissociation for $(C=C)$ and $(\mathrm{C-C})$ at $298\ \mathrm{K}$ are $+590$ and $+331\ \mathrm{kJ\ mol^{-1}}$, respectively. The enthalpy of polymerization is $-360\ \mathrm{kJ\ mol^{-1}}$. Find the value of n.

(A) Submit

Q.50

A solution containing 8.0 g of nicotine in 92 g of water freezes 0.925 degrees below the normal freezing point of water. If the molal freezing point depression constant, $\mathrm{K_f} = 1.85^{\circ}\mathrm{C\ mol^{-1}}$, then the molar mass of nicotine is: (mark answer to nearest integer)

(A) Submit

Q.51

A locker can be opened by dialing a fixed three digit code (between 000 and 999). A stranger who does not know the code tries to open the locker by dialing three digits at random. The probability that the stranger succeeds at the kth trial is

(A) $\frac{k}{999}$

(B) $\frac{k}{1000}$

(C) $\frac{k-1}{1000}$

(D) None of these

Q.52

If N is the number of positive integral solutions of the equation $x_1x_2x_3x_4 = 770$, then the value of N is

(A) 250

(B) 252

(C) 254

(D) 256

Q.53

The value of the determinant $\begin{vmatrix} \sqrt{13}+\sqrt{3} & 2\sqrt{5} & \sqrt{5} \\ \sqrt{15}+\sqrt{26} & 5 & \sqrt{10} \\ 3+\sqrt{65} & \sqrt{15} & 5 \end{vmatrix}$ is

(A) $5(\sqrt{6}-5)$

(B) $5\sqrt{3}(\sqrt{6}-5)$

(C) $\sqrt{5}(\sqrt{6}-\sqrt{3})$

(D) $\sqrt{2}(\sqrt{7}-\sqrt{5})$

Q.54

The sum of the roots of the equation $2^{33x-2}+2^{11x+2}=2^{22x+1}+1$ is

(A) $\frac{1}{11}$

(B) $\frac{2}{11}$

(C) $\frac{3}{11}$

(D) $\frac{4}{11}$

Q.55

The fractional part of $\frac{2^{78}}{31}$ is

(A) $\frac{2}{31}$

(B) $\frac{4}{31}$

(C) $\frac{6}{31}$

(D) $\frac{8}{31}$

Q.56

Let $P=(1,0)$ and $Q=(-1,0)$ are two fixed points and R is a variable point on one side of the line PQ such that $\angle RPQ-\angle RQP=45^\circ$, then the locus of R is

(A) $y^2-x^2+2xy-1=0$

(B) $x^2-y^2+2xy+1=0$

(C) $y^2+x^2-2xy=1$

(D) $y^2 - x^2 - 2xy + 1 = 0$

Q.57

The solution to the differential equation $\frac{dy}{dx} = \frac{yf'(x) - y^2}{f(x)}$ where $f(x)$ is a given function is

(A) $f(x) = y(x + c)$

(B) $f(x) = cxy$

(C) $f(x) = c(x + y)$

(D) $yf(x) = cx$

Q.58

The number of ways of arranging the letters AAAAA, BBB, CCC, D, EE and F in a row, if the letters B are separated from one another, is equal to

(A) $\frac{13!}{5!3!3!2!}$

(B) $\frac{14!}{3!3!2!}$

(C) $\frac{15!}{(3!)^2 2!5!}$

(D) ${}^{13}C_3 \times \frac{12!}{5!3!2!}$

Q.59

Let $\alpha,\ \beta$ and γ be the roots of $f(x) = 0$, where $f(x) = x^3 + x^2 - 5x - 1$. Then the value of $[\alpha] + [\beta] + [\gamma]$ is equal to (where $[\cdot]$ denotes the greatest integer function)

(A) -1

(B) 1

(C) 4

(D) -3

Q.60

$A = \begin{bmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \\ l_3 & m_3 & n_3 \end{bmatrix}$ and $B = \begin{bmatrix} p_1 & q_1 & r_1 \\ p_2 & q_2 & r_2 \\ p_3 & q_3 & r_3 \end{bmatrix}$ p_i , q_i , r_i are the cofactors of the elements $l_i,\ m_i,\ n_i$ for $i = 1$,2, 3. If $(l_1, m_1, n_1), (l_2, m_2, n_2)$ and (l_3, m_3, n_3) are the direction cosines of three mutually perpendicular lines, then $(p_1, q_1, r_1), (p_2, q_2, r_2)$ and (p_3, q_3, r_3) are

(A) the direction cosines of three mutually perpendicular lines

(B) the direction ratios of three mutually perpendicular lines which are not direction cosines

(C) the direction cosines of three lines which need not be perpendicular

(D) the direction ratios but not the direction cosines of three lines which need not be perpendicular

Q.61

If $\int \frac{1-2019\cos^2 x}{(\sin x)^{2019}\cos^2 x}dx = \tan x \cdot f\left(x\right) + C$ where 'C' is an integral constant then $f\left(\frac{\pi}{2}\right) + f'\left(\frac{\pi}{2}\right) + f\left(-\frac{\pi}{2}\right)$ is equal to

(A) 3

(B) 1

(C) 0

(D) -1

Q.62

Universal set,
$U = \left\{x | x^5 - 6x^4 + 11x^3 - 6x^2 = 0\right\}$
$A = \left\{x | x^2 - 5x + 6 = 0\right\}$
$B = \left\{x | x^2 - 3x + 2 = 0\right\}$
what is $(A \cap B)'$ equal to?

(A) $\{1, 3\}$

(B) $\{1, 2, 3\}$

(C) $\{0, 1, 3\}$

(D) $\{0, 1, 2, 3\}$

Q.63

Let $f(x)$ be a function which is differentiable any number of times and $f\left(2x^2 - 1\right) = 2x^3 f\left(x\right),\ \forall\ x \in R$. Then $f^{(2010)}\left(0\right) =$
(Here, $f^{(n)}(x) = n^{\text{th}}$ order derivative of $f(x)$)

(A) -1

(B) 1

(C) 0

(D) data is insufficient

Q.64

The area bounded by $\mathrm{y} = f(\mathrm{x})$, x-axis, y-axis and the line $\mathrm{y} = 1$ whose $f(\mathrm{x}) = 1 + \frac{1}{x}\int_1^x f(t)dt, x > 0$ is

(A) $2(\mathrm{e} + 1)$

(B) $2\left(1 - \frac{1}{e}\right)$

(C) $2(\mathrm{e} - 1)$

(D) $\left(1 - \frac{1}{e}\right)$

Q.65

The axis of a parabola lie along the line $y = x$ and the distance of its vertex from origin is $\sqrt{2}$ and that of focus is $2\sqrt{2}$. If both focus and vertex lie in the first quadrant, then the equation of the parabola will be

(A) $(x+y)^2 = (x-y-2)$

(B) $(x-y)^2 = (x+y-2)$

(C) $(x-y)^2 = 4(x+y-2)$

(D) $(x-y)^2 = 8(x+y-2)$

Q.66

Choose the correct option

(A) $|\vec{a}| = |\vec{b}| = |\vec{a} - \vec{b}| = 1$ then angle between $\vec{a}$ and $\vec{b}$ is $\pi/2$.

(B) $\frac{\vec{a}+\vec{b}}{2}$ is a vector in the direction of angle bisector of vectors $\vec{a}$ and $\vec{b}$.

(C) $(\vec{a} \cdot \hat{i})^2 + (\vec{a} \cdot \hat{j})^2 + (\vec{a} \cdot \hat{k})^2 = a^2$.

(D) Vector perpendicular to $-\hat{i} + \hat{j} + \hat{k}$ and coplanar with $\hat{i} + \hat{j} + \hat{k}$ and $-\hat{i} + \hat{j} + \hat{k}$ is $3\hat{i} + 2\hat{j} + \hat{k}$.

Q.67

If the function $f(x) = Pe^{2x} + Qe^{x} + Rx$ satisfies the conditions $\mathrm{f}(0) = -1, \mathrm{f}'(\log 2) = 31$ and $\int_0^{\log 4} (f(x) - Rx)dx = \frac{39}{2}$, then

(A) $P = 5, Q = -6, R = 3$

(B) $P = 5, Q = 2, R = 5$

(C) $\mathrm{P} = -5, \mathrm{Q} = 6, \mathrm{R} = 3$

(D) $P = 3, Q = 2, R = 3$

Q.68

If $y = \tan^{-1}\frac{1}{1+x+x^2} + \tan^{-1}\frac{1}{x^2+3x+3} + \tan^{-1}\frac{1}{x^2+5x+7} + \ldots.+$ upto $2n$ terms $(\forall x \geq 0)$, then $y(0)$ is

(A) $\tan^{-1}(n)$

(B) $\tan^{-1}(2n)$

(C) $2\tan^{-1}(n)$

(D) 0

Q.69

Let $a_1 = 0$ and $a_1, a_2, a_3, \ldots\ldots, a_n$ be real numbers such that $|a_i| = |a_{i-1} + 1|$ for all i, then the A.M. of the numbers $a_1, a_2, a_3, \ldots\ldots, a_n$ has the value A where :

(A) $A < -\frac{1}{2}$

(B) $A < -1$

(C) $A \geq -\frac{1}{2}$

(D) $A = -2$

Q.70

If $f : [1, \infty) \to [\sqrt{2}, \infty)$ be a function defined by $f(x) = \sqrt{x^2 - 2x + 3}$ and g be inverse function of f then derivative of $g\left(f^2(x)\right)$ at $x = 3$ is:

(A) $\frac{6}{\sqrt{34}}$

(B) $\frac{\sqrt{34}}{6}$

(C) $\frac{12}{\sqrt{34}}$

(D) $\frac{24}{\sqrt{34}}$

Q.71

If $\lim\limits_{x \to 0^+} \frac{\{x\}^{\{x\}} \cos\left(\{x\}^{\{\sin x\}+1}\right)}{\cos(\tan\{x\} - x)\cos(\{x+1\} - x)} = k$, then k is equal to (where $\{.\}$ denotes fractional part function)

(A) Submit

Q.72

Given two skew-lines with parametric equations $x = 1 + t,\ y = 1 + 6t,\ z = 2t$ and
$x = 1 + 2s,\ y = 5 + 15s,\ z = -2 + 6$s. The shortest distance between these two given lines is equal to

(A) Submit

Q.73

For any integer k, let $\alpha_k = \cos\left(\frac{k\pi}{7}\right) + i\sin\left(\frac{k\pi}{7}\right)$, where $i = \sqrt{-1}$. The value of the expression $\frac{\sum_{k=1}^{12} |\alpha_{k+1} - \alpha_k|}{\sum_{k=1}^{3} |\alpha_{4k-1} - \alpha_{4k-2}|}$ is

(A) Submit

Q.74

Let $X_1,\ X_2,\ X_3 \ldots$ are in arithmetic progression with a common difference equal to d which is a two digit natural number. $y_1,\ y_2,\ y_3 \ldots$ are in geometric progression with common ratio equal to 16 . Arithmetic mean of $X_1, X_2 \ldots X_n$ is equal to the arithmetic mean of $y_1, y_2 \ldots y_n$ which is equal to 5. If the arithmetic mean of $X_6, X_7 \ldots X_{n+5}$ is equal to the arithmetic mean of $y_{\mathrm{P}+1}, y_{\mathrm{P}+2} \ldots y_{\mathrm{P}+n}$, then d is equal to

(A) Submit

Q.75

If $4\cos 36^\circ + \cot\left(7\frac{1}{2}^\circ\right) = \sqrt{n_1} + \sqrt{n_2} + \sqrt{n_3} + \sqrt{n_4} + \sqrt{n_5} + \sqrt{n_6}$, then the value of $\left(\frac{\sum_{i=1}^{6} n_i^2}{7}\right)$ is equal to

(A) Submit

Answers & Solutions

Q.1 Answer:

—

Solution:

As, $\left[\mathrm{MLT}^{-2}\right] = \left[\mathrm{L}^{2a}\right]\left[\mathrm{L}^{b}\,\mathrm{T}^{-b}\right]\left[\mathrm{M}^{c}\,\mathrm{L}^{-3c}\right] = \left[\mathrm{M}^{c}\,\mathrm{L}^{2a+c-3c}\,\mathrm{T}^{-b}\right]$

Comparing powers of M, L and T, we get

$c = 1, 2a + b - 3c = 1, -b = -2$ or $b = 2$

$2a + 2 - 3(1) = 1$

$\Rightarrow 2a = 2$

or $a = 1$

Q.2 Answer:

$W_1 < W_2 < W_3$

Solution:

Voltage across B_3 is greatest, hence B_3 will show maximum brightness. In series combination of bulbs, the bulb of lesser wattage will glow more bright.

Hence $W_2 > W_1$. So, $W_1 < W_2 < W_3$.

Q.3 Answer:

20 cm

Solution:

Excess pressure $\frac{4S}{r_1} - \frac{4S}{r_2} = \frac{4S}{r}$

or $\frac{1}{r} = \frac{1}{r_1} - \frac{1}{r_2} = \frac{1}{4} - \frac{1}{5} = \frac{1}{20}$ or $r = 20$ cm

Q.4 Answer:

pV

Solution:

Work done during the complete cycle

= Area $ABCDA = AD \times AB = p \times V = pV$

Q.5 Answer:

all 3

Solution:

$Q_{(i)} = (M_{Br} - M_{Kr})\, c^2 > 0$

$Q_{(ii)} = (M_{Br} - M_{Se} - 2M_e)\, c^2 > 0$

$Q_{(ii)} = (M_{Br} - M_{Se})\, c^2 > 0$

Hence all the nuclear reactions are possible.

Q.6 Answer:

If both Assertion and Reason are true and Reason is the correct explanation of Assertion.

Solution:

As the electric field is uniform, the magnitude of electric field is equal i.e., E at every point within the field.

In a dipole, the charges have same charge magnitude Q but opposite polarity.

As the net charge on dipole, $Q_{\text{net}} = 0$

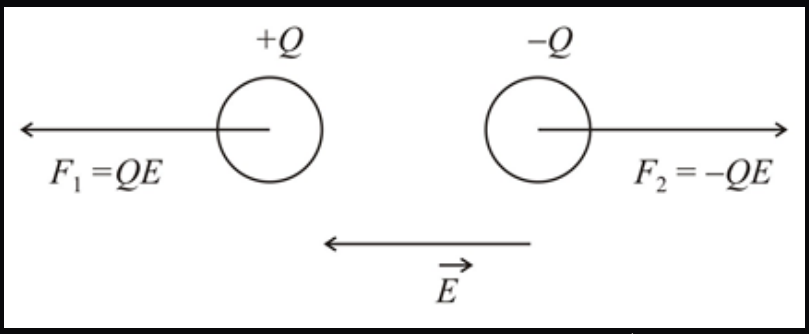
Force on a charge Q in an electric field E is

$F = QE$

Net Force on the dipole in uniform electric field,

$F_{\text{net}} = EQ - EQ$

$\Rightarrow F_{\text{net}} = 0$



Thus, work done on dipole for displacement $\vec{d}$,

$W = \vec{F_1} \cdot \vec{d} + \vec{F_2} \cdot \vec{d} = \left(\vec{F_1} + \vec{F_2}\right) \cdot \vec{d}$

$\Rightarrow W = \vec{F}_{\text{net}} \cdot \vec{d} = 0 \times d$

$\Rightarrow W = 0$

Q.7 Answer:

1 and 2

Solution:

Magnetic field strength in a solenoid is given by $B = \mu_0 ni$

where, μ_0 = permeability of free space,

n = number of turns per unit length and i = current flowing through solenoid.

So, it is clear that magnetic field strength B in a solenoid depends on n and i only, it does not depends on diameter of solenoid. Therefore, statements 1 and 2 are correct.

Q.8 Answer:

$0.2\ \text{mm}$

Solution:

$20\text{VSD} = 16\,\text{MSD}; \quad 1\text{MSD} = 1\ \text{mm}$
$1\text{VSD} = \frac{16}{20} \times 1\,\text{mm} = 0.8\ \text{mm}$
$\text{LC} = 1\text{MSD} - 1\text{VSD} = 1 - 0.8 = 0.2\ \text{mm}$

Q.9 Answer:

The potential difference across the plates increases

Solution:

Capacitance C of a parallel plate air capacitor of plate area A and plate separation d is given by
$C = \frac{\varepsilon_0 A}{d} \quad \ldots . (1)$
When it is charged with the help of a source of potential difference V, then suppose the charge on it be q. If the charging battery is disconnected and separation d is increased, then charge q on the capacitor remains constant.
Electric field between the plates is given by
$E = \frac{\sigma}{\varepsilon_0} = \frac{q}{A\varepsilon_0} \quad \ldots . (2)$
As charge remains constant, electric field also remains constant.
Also, from equation (1), as d increases, capacitance C decreases.
Potential difference across the plates is given by
$V = Ed \quad \ldots . (3)$
$\therefore$ **As d increases, V also increases.**

Q.10 Answer:

$\frac{2\text{h}}{\pi}$

Solution:

$10 = \frac{\text{n(n+1)}}{2} \Rightarrow \text{n} = 4$
$\text{mvr} = \frac{4 \times \text{h}}{2\pi} = \frac{2\,\text{h}}{\pi}$

Q.11 Answer:

$\frac{3}{2}\sqrt{\frac{5\text{GM}}{a}}$

Solution:

Let there are two stars 1 and 2
Let P is a point between C_1 and C_2, where gravitational field strength is zero. Or at P field strength due to star 1 is equal and opposite to the field strength due to star 2 . Hence,
$\frac{\text{GM}}{\text{r}_1^2} = \frac{\text{G(16M)}}{\text{r}_2^2}$
or $\frac{\text{r}_2}{\text{r}_1} = 4$
also $r_1 + r_2 = 10a$
$\therefore r_2 = \left(\frac{4}{4+1}\right)(10a) = 8a$

and $r_1 = 2a$

Now, the body of mass m is projected from the surface of larger star towards the smaller one. Between C_2 and P it is attracted towards 2 and between C_1 and P it will be attracted towards 1 . Therefore, the body should be projected to just cross point P because beyond that the particle is attracted towards the smaller star itself.

From conservation of mechanical energy $\frac{1}{2}mv_{min}^2$

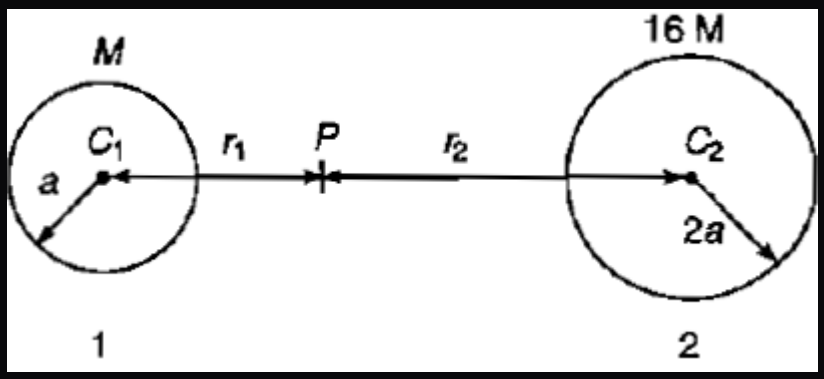
= Potential energy of the body at P-Potential energy at the surface of the larger star.

$$\therefore \frac{1}{2}mv_{min}^2 = \left[-\frac{GMm}{r_1} - \frac{16GMm}{r_2}\right] - \left[-\frac{GMm}{10a-2a} - \frac{16GMm}{2a}\right]$$

$$= \left[-\frac{GMm}{2a} - \frac{16GMm}{8a}\right] - \left[-\frac{GMm}{8a} - \frac{8GMm}{a}\right]$$

$$\text{or } \frac{1}{2}mv_{min}^2 = \left(\frac{45}{8}\right)\frac{GMm}{a}$$

$$\therefore v_{min} = \frac{3\sqrt{5}}{2}\left(\sqrt{\frac{GM}{a}}\right)$$



Q.12 Answer:

$\frac{37}{7}$ m

Solution:

Using equation of trajectory:

$y = x\tan\alpha\left(1 - \frac{x}{R}\right)$;

For points $(3\ ,\ 4)$

$\Rightarrow\ 4 = 3\tan\alpha\left(1 - \frac{3}{R}\right)$ ...(i)

For point $(4\ ,\ 3)$

$\Rightarrow\ 3 = 4\tan\alpha\left(1 - \frac{4}{R}\right)$ ...(ii)

Dividing (i) and (ii)

$\Rightarrow\ \frac{4}{3} = \frac{3}{4}\frac{\left(1-\frac{3}{R}\right)}{1-\frac{4}{R}} \Rightarrow 16\left(1 - \frac{4}{R}\right) = 9\left(1 - \frac{3}{R}\right)$

$\Rightarrow\ 7 = \frac{64}{R} - \frac{27}{R} \Rightarrow R = \frac{37}{7}$ m

Q.13 Answer:

$\frac{Ml^2}{6}$

Solution:

$l_1 = \frac{Ml^2}{12}, \quad I_2 = \frac{M^2}{12}$

Here, there is no use of α because the moment of inertia depends on the perpendicular distance of the particles from the axis of rotation.

So, $I_{\text{net}} = I_1 + I_2 = \frac{Ml^2}{12} + \frac{Ml^2}{12} = \frac{Ml^2}{6}$

$l_1 = \frac{Ml^2}{12}, \quad I_2 = \frac{M^2}{12}$

Here, there is no use of α because the moment of inertia depends on the perpendicular distance of the particles from the axis of rotation.

So, $I_{\text{net}} = I_1 + I_2 = \frac{Ml^2}{12} + \frac{Ml^2}{12} = \frac{Ml^2}{6}$

Q.14 Answer:

(a) (b) (c) (d)

(i) (ii) (iii) (iv)

Solution:

Infrared - Muscular treatment

Radio wave - Broadcasting

X Rays - To detect fracture

Uv Rays $\rightarrow$ Absorbed ozone layer

Q.15 Answer:

$\frac{d^2}{9D}$

Solution:

For dark fringe,

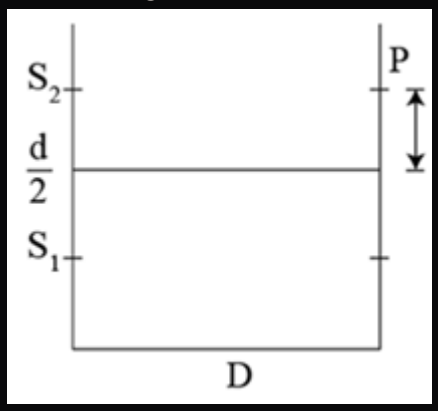
$\frac{\mathrm{Yd}}{\mathrm{D}} = (2\,\mathrm{m} - 1)\frac{\lambda}{2}$

Here, $m = 5, Y = \frac{d}{2}$

$\therefore \frac{d}{2}\frac{d}{D} = (2 \times 5 - 1)\frac{\lambda}{2}$

$\mathrm{Or}\ \frac{d^2}{D} = 9\lambda$

$\therefore \lambda = \frac{d^2}{9D}$



y = $\frac{d}{2}$

Q.16 Answer:

OR gate

Solution:

Truth table for given combination is

A	B	$\overline{A}$	$\overline{B}$	$\overline{A}\cdot\overline{B}$	$X=\overline{\overline{A}\cdot\overline{B}}$
0	0	1	1	1	0
0	1	1	0	0	1
1	0	0	1	0	1
1	1	0	0	0	1

This is same as that of OR gate.

Q.17 Answer:

$\frac{2M}{\pi}$

Solution:

Let, the pole strength of wire $= m$

Magnetic moment =Pole strength $\times$Separation between poles.

For straight wire:

Magnetic moment, $M = ml$

When a wire of length l is bent in semi-circular shape, then, $l = \frac{\pi d}{2}$.

where, d is the diameter of a semi-circle.

$\Rightarrow d = \frac{2l}{\pi}$

Also in semi-circle, the separation between poles is its diameter.

Hence, the magnetic moment for semi-circle, $M' = m\left(\frac{2l}{\pi}\right) = \frac{2M}{\pi}$

Q.18 Answer:

$\frac{3}{2}MB$

Solution:

KE will be maximum when PE is minimum, and

PE is minimum at $\theta = 0^\circ$

Applying energy conservation at $\theta = 120^\circ$ and $\theta = 0^\circ$

$-MB\cos 120^\circ + 0 = -MB\cos 0^\circ + \mathrm{KE}_{\max}$

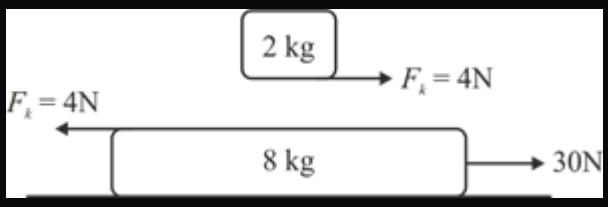
$\Rightarrow \mathrm{KE}_{\max} = \frac{3}{2}MB$

Q.19 Answer:

$\sqrt{3}s$

Solution:

Free body diagram,



For $8\ \text{kg}, \text{a}_1 = \frac{30-4}{8} = \frac{26}{8} = 3.25\ \text{m/s}^2$

For $2\ \text{kg},$

$a_2 = \frac{F_k}{2} = \frac{4}{2} = 2\ \text{m/s}^2$

Relative acceleration of upper block with respect to lower block

$$a_{21} = a_2 - a_1$$
$$= 2 - 3.25$$
$$= -1.25\ \text{m/s}^2\ (\text{backwards})$$
$$\Rightarrow \text{u}_{21} = 0$$
$$\Rightarrow \text{s}_{21} = -2\ \text{m}\ (\text{backwards})$$

From equation of motion,

$$\text{s}_{21} = \text{u}_{21}t + \frac{1}{2}\text{a}_{21}t^2$$
$$-2 = 0 + \frac{1}{2}(-1.25) \times t^2$$
$$\frac{4}{1.25} = t^2$$
$$3 = t^2 \quad (\text{approx})$$
$$t = \sqrt{3}\ \text{s}$$

Q.20 Answer:

$\frac{Q}{16}$

Solution:

The rate of heat flow is given by

$\frac{Q}{t} = \text{K.A.}\frac{\Delta T}{\ell}$

Area of Original rod $\text{A} = \pi R^2;$

Areal of new rod $\text{A}' = \frac{\pi R^2}{4}.$

Volume of original rod will be equal to the volume of new rod.

$\therefore \pi R^2 \ell = \pi \left(\frac{R}{2}\right)^2 \ell'$

$\Rightarrow \frac{\ell'}{\ell} = \frac{R^2}{\left(\frac{R^2}{4}\right)} = 4$

$\therefore \frac{Q'}{Q} = \frac{A'}{A}\frac{\ell}{\ell'} = \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16}$

$\therefore Q' = \frac{Q}{16}$

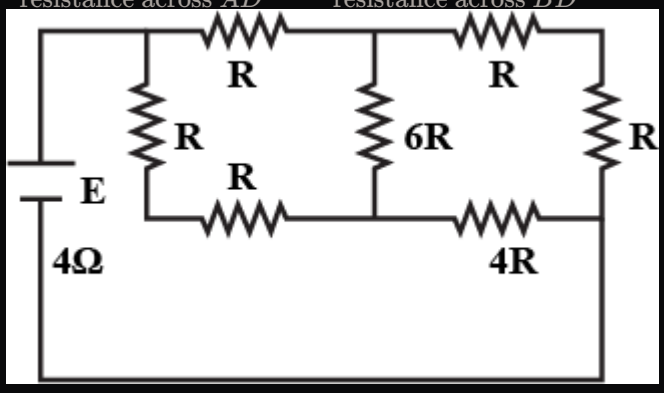
Q.21 Answer:

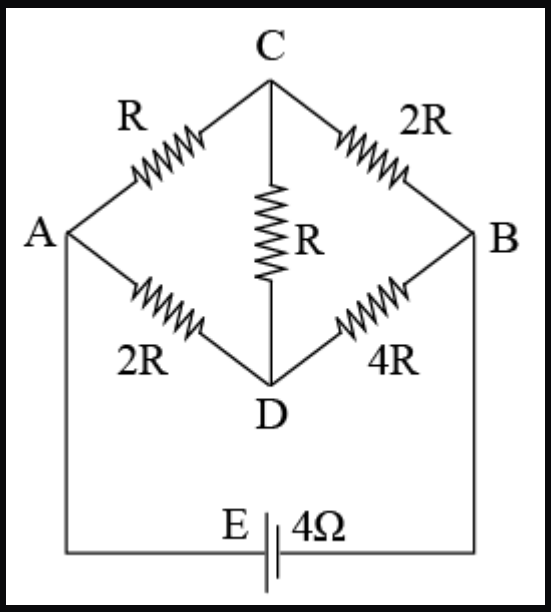
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Solution:

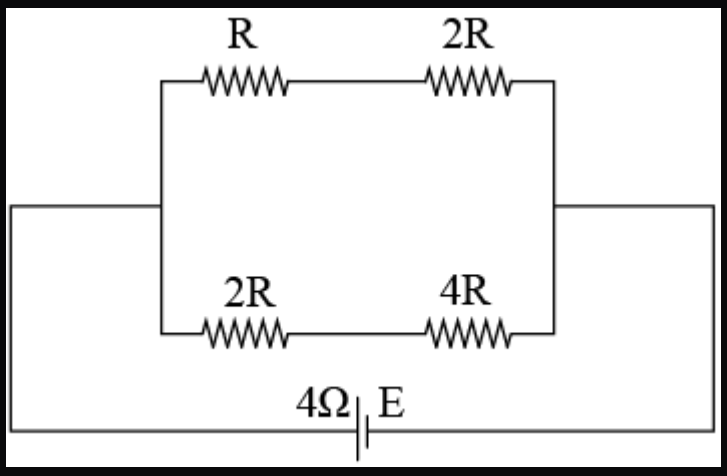
The given circuit is that of a Wheatstone bridge. The circuit is balanced one since,

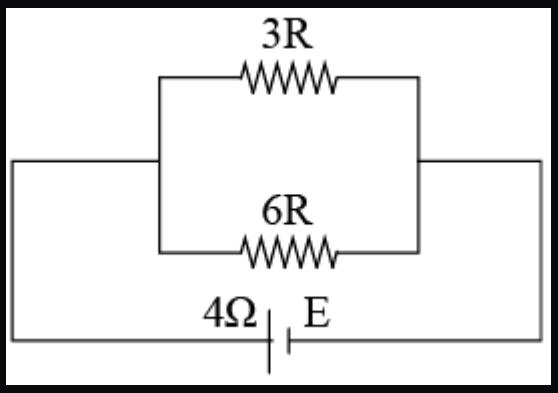
$$\frac{\text{resistance across } AC}{\text{resistance across } AD} = \frac{\text{resistance across } CB}{\text{resistance across } BD}$$

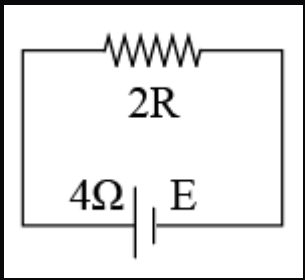




Thus, no current will flow across $6R$ of the side CD. The given circuit will now be equivalent to For maximum power net external resistance







= Total internal resistance

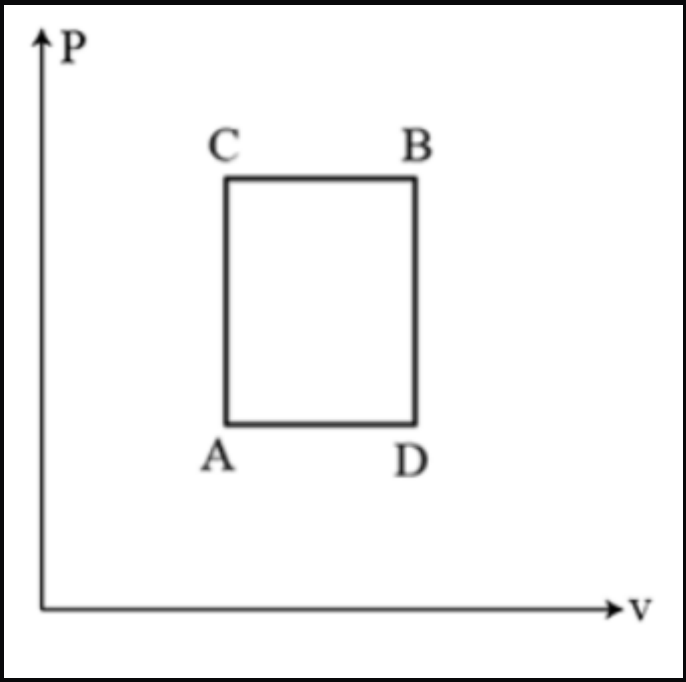
or $2R = 4$

or $R = 2\Omega$

Q.22 Answer:

—

Solution:



Δ Uremains same for both paths ACB and ADB

$\Delta Q_{ACB} = \Delta W_{ACB} + \Delta U_{ACB}$

$\Rightarrow 60\ J = 30\ J + \Delta U_{ACB}$

$\Rightarrow U_{ACB} = 30\ J$

$\therefore \Delta U_{ADB} = \Delta U_{ACB} = 30\ J$

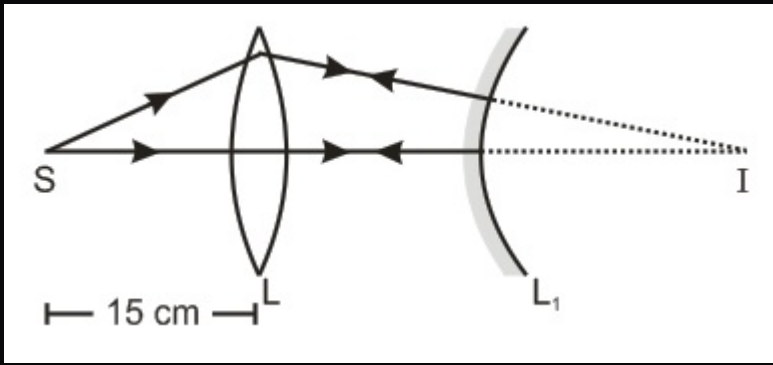
$\Delta Q_{ADB} = \Delta U_{ADB} + \Delta W_{ADB}$

$= 10\ J + 30\ J = 40\ J$

Q.23 Answer:

—

Solution:



The equation for the lens is

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}. \qquad \text{.....(i)}$$

Here u = -15 cm and f = + 10 cm.

Using equation (i),

$$\frac{1}{v} + \frac{1}{15\ \text{cm}} = \frac{1}{10\ \text{cm}}$$

$$\frac{1}{v} = \frac{1}{10\ \text{cm}} - \frac{1}{15\ \text{cm}} = \frac{1}{30\ \text{cm}}$$

or, $v = 30\ \text{cm}.$

The positive sign of *v* shows that the image I_1 is formed to the right of the lens in the figure. The diverging mirror is to be placed to the right in such a way that the light rays fall on the mirror perpendicularly. Then only the rays will retrace their path and form the final image on the object. Thus, the image I_1 formed by the lens should be at the centre of curvature of the mirror.

We have, LI_1 = 30 cm,

MI_1 = R = 2f = 24 cm,

Hence, LM = LI - MI_1 = 6 cm.

Thus, the mirror should be placed 6 cm to the right of the lens.

Q.24 Answer:

—

Solution:

According to Gauss's law of electrostatics,

$$\oint \vec{E} \cdot \mathrm{d}\vec{A} = \frac{1}{\varepsilon_0} \int (\rho \mathrm{d}V) = \frac{1}{\varepsilon_0} \int_0^r kr^a \times 4\pi r^2 \mathrm{d}r \Rightarrow E \times 4\pi R^2 = \left(\frac{4\pi k}{\varepsilon_0}\right) \frac{r^{(a+3)}}{a+3}$$

At $r = R$, $E_1 = \frac{kR^{(a+1)}}{\varepsilon_0(a+3)}$

At $r = \frac{R}{2}$, $E_2 = \frac{k}{\varepsilon_0(a+3)} \left(\frac{R}{2}\right)^{a+1}$

Given, $E_2 = \frac{1}{8} E_1$

$$\Rightarrow \frac{k}{\varepsilon_0(a+3)} \left(\frac{R}{2}\right)^{a+1} = \frac{1}{8} \times \frac{kR^{(a+1)}}{\varepsilon_0(a+3)} \Rightarrow 2^{1/(a+1)} = \frac{1}{8} \Rightarrow a = 2$$

Q.25 Answer:

—

Solution:

$$\frac{1}{2} k A^2 = \frac{p^2}{2m}$$
$$\Rightarrow \left(\frac{A_1}{A_2}\right)^2 = \frac{m_2}{m_1} = \frac{1024}{900}$$
$$\Rightarrow \frac{A_1}{A_2} = \frac{32}{30} = \frac{16}{15} = \frac{16}{16-1}$$
$$\therefore \alpha = 16$$

Q.26 Answer:

Mn^{2+}

Solution:

In Mn^{2+} number of unpaired $de^- = 5$. So it has maximum magnetic moment according to the formula.

$\mu = \sqrt{n(n+2)}$

$Ti^{3+} \rightarrow 3d^1 4s^0;$

$Sc^{3+} \rightarrow 3d^0;$

$Mn^{2+} \rightarrow 3d^5 4s^0;$

$Zn^{2+} \rightarrow 3d^{10} 4s^0$

Q.27 Answer:

0.0172 M

Solution:

$(As_2^{3+} \rightarrow 2As^{5+} + 4e^-)$

$(Mn^{7+} + 5e^- \rightarrow Mn^{2+})$

Meq. of $As_2O_3 =$ Meq of $KMnO_4$

$$\frac{0.1156}{198/4} \times 1000 = M \times 5 \times 27.08$$

$$\therefore M = 0.0172$$

$$\therefore M_{MnO_4^-} = 0.0172\ M$$

Q.28 Answer:

Both (a) and (c)

Solution:

The Lewis structures of a substance having resonance are termed as resonating structures.

All the resonating structures of a substance have identical position of atoms and the same number of electron pairs but these may differ in the number and position of the pi-bonds and the location of the charges.

Q.29 Answer:

Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

Solution:

Due to inert pair effect Pb has four electrons in its valence shell but it shows $+2$ oxidation state. In other words due to inert pair effect $+2$ oxidation state is more stable than $+4$ of Pb.
So, both Assertion and Reason are true and Reason is the correct explanation of Assertion.

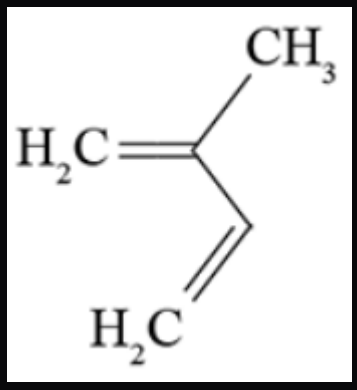
Q.30 Answer:

(p)-4, (q)-3, (r)-2, (s)-1

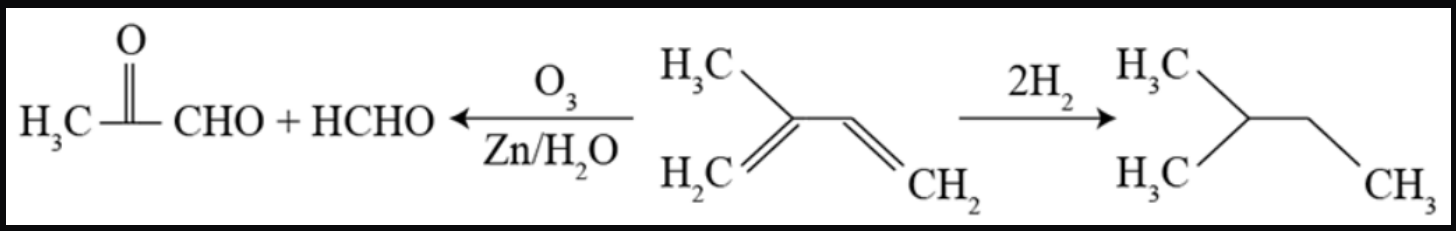
Solution:

Nitrobenzene on reduction with $\mathrm{Sn/HCl}$ gives aniline. With Zn it gives phenyl hydroxylamine. With methanolic NaOMe nitrobenzene gives Azoxybenzene. On reaction with Zn and strong base it gives Hydrazobenzene.

Q.31 Answer:



Solution:



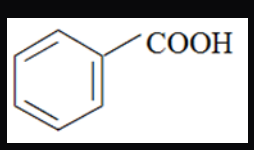
Q.32 Answer:

$\mathrm{O_2^{2-}}$

Solution:

$\mathrm{O_2^{2-}}$ has no unpaired electron, hence it is diamagnetic.
$\mathrm{O_2^{2-}} = \sigma 1s^2 \sigma s^{2*} \sigma 2s^{2*} \sigma 2s^2 \sigma 2p_x^2 \pi 2p_y^2 \pi 2p_z^{2*} \pi p_y^{2*} \pi 2p_z^2$

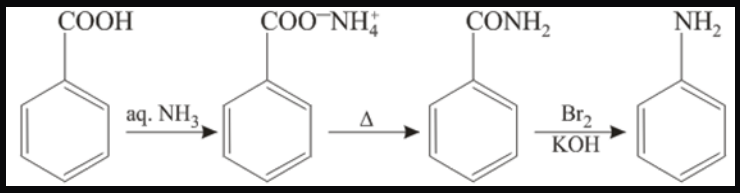
Q.33 Answer:



Solution:

$C_7H_6O_2(P) \xrightarrow[(2)\Delta]{(1)aq.NH_3} Q \xrightarrow[KOH]{Br_2} R$

Here P is an acid, Q is amide and Q to R is Hoffmann's bromamide degradation as follows



Q.34 Answer:

Al^{3+} has high nuclear charge than Mg^{2+}

Solution:

Atomic radius is inversely proportional to nuclear attraction. Al^{3+} has high nuclear charge than Mg^{2+} therefore Al^{3+} has low ionic radius than Mg^{2+}

Q.35 Answer:

$56\ mL$

Solution:

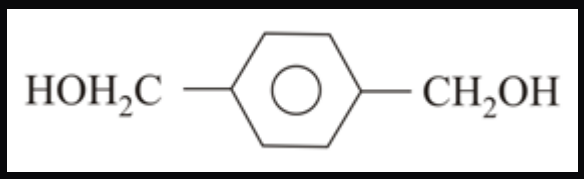
$W = \frac{i \times t \times E}{F}$

$= \frac{5 \times 193 \times 8}{96500}$

$\because$ **At** NTP, **volume of** $32 g O_2 = 22400\ mL$

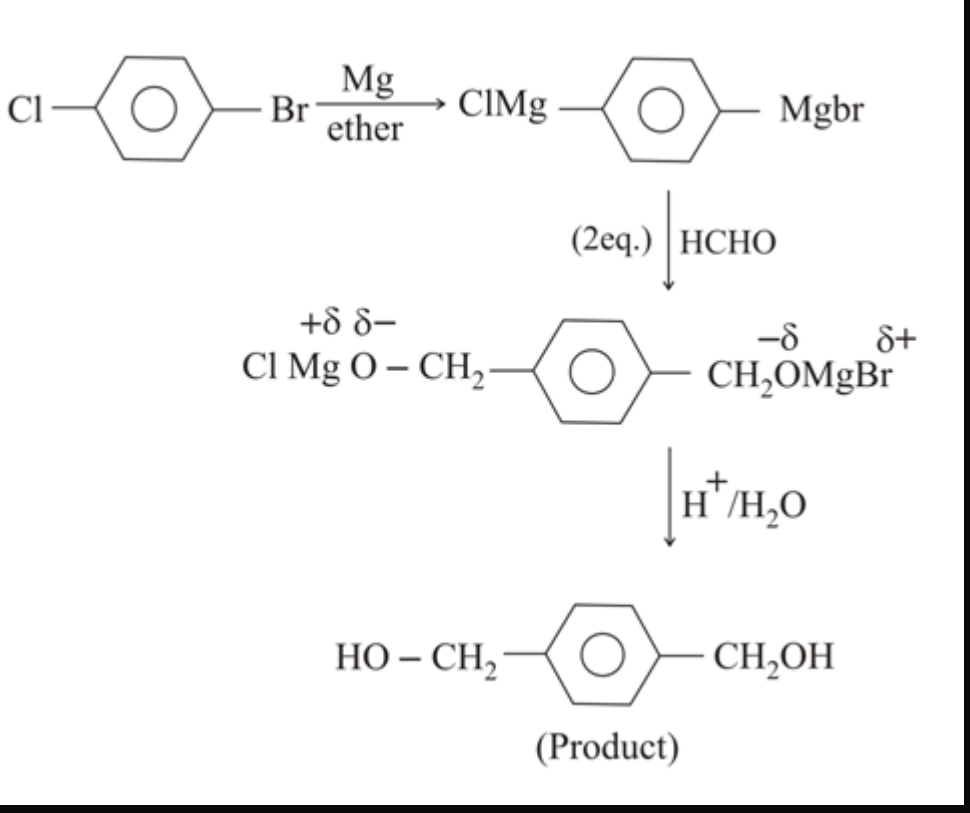
$\therefore$ **Volume of** $0.08 g O_2 = \frac{22400 \times 0.08}{32} = 56\ mL$

Q.36 Answer:



Solution:

The given reaction can be completed as follows:



Q.37 Answer:

Yellow

Solution:

In transition metals, one of the colours of the visible spectrum is absorbed and the transition metal ion gets the complementary colour. In this case if violet is absorbed, its complementary colour is Yellow.
So, When the transition element absorbs violet light, it emits yellow light.

Q.38 Answer:

If both assertion and reason are true but reason is not the correct explanation of assertion.

Solution:

Sulphonation of phenol is a reversible process.
At lower temperatures, due to thermodynamic control and formation of intramolecular hydrogen bonding ortho hydroxy benzene is stable.
At high temperatures, due to kinetic control, para hydroxy benzene is stable.
Therefore, Both assertion and reason are true but reason is not the correct explanation of assertion.

Q.39 Answer:

3

Solution:

In this sequence of reaction three times aldol reaction occurs first than cross Cannizaro reaction takes place as follows

$$CH_3 - CHO + 3HCHO \xrightarrow{OH^-} (CH_2OH)_3 - C - CHO$$

$$(CH_2OH)_3 - C - CHO \xrightarrow{OH^-} C - (CH_2OH)_4$$

Q.40 Answer:

II < III < IV < I < V

Solution:

$CO_3^{2-} \rightarrow sp^2 \rightarrow \%s = 33.33\%$

$XeF_4 \rightarrow sp^3d^2 \rightarrow \%s = 16.67\%$

$I_3^- \rightarrow sp^3d \rightarrow \%\ s = 20\%$

$NCl_3 \rightarrow sp^3 \rightarrow \%\ s = 25\%$

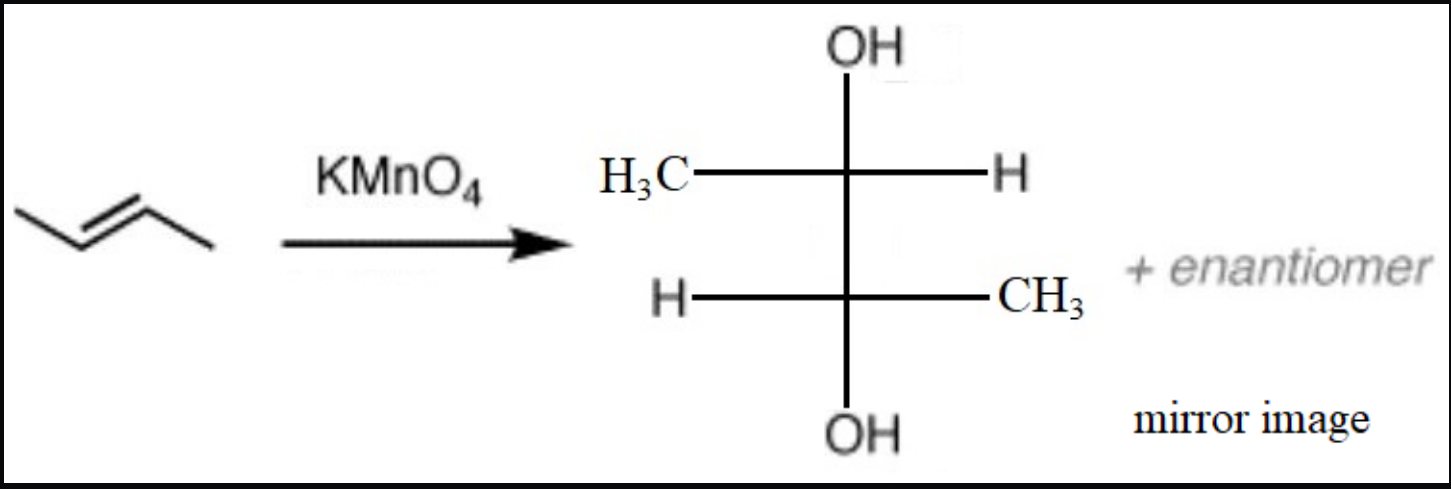
$B_eCl_2 \rightarrow sp \rightarrow \%\ s = 50\%$

Q.41 Answer:

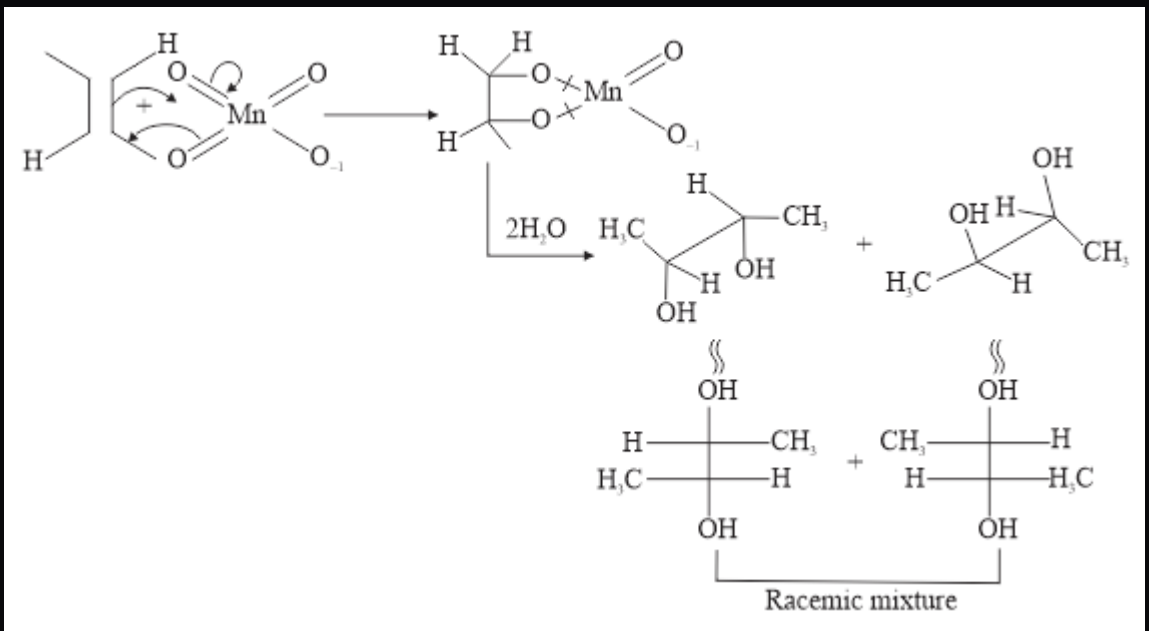
A is a racemic mixture of d and l-2, 3-butanediol formed by syn addition

Solution:

Alkenes form diol, when treated with alkaline $KMnO_4$. The reaction occurs by syn-addition



Reaction follows syn addition



Q.42 Answer:

5'-UCGAUGUGA-3'

Solution:

The two stands of DNA are complementary to each other and the bases of one stand are paired with the bases of other strand by means of hydrogen bonding, i.e., $A = T$ and $G \equiv C$. But uracil (U) instead of thymine (T) is present in case of RNA. Thus, the sequence of base on m-RNA molecule is
DNA strand 5'-AGCTACACT-3 m-RNA 5'-UCGAUGUCA-3

Q.43 Answer:

10.5

Solution:

$\mathrm{Fe_{0.95}O_1}$

Let there are x ions of $\mathrm{Fe^{3+}}$ then $(0.95 - \mathrm{x})$ ions will be $\mathrm{Fe^{2+}}$.

Compound is electrically neutral. $+3x + 2(0.95 - x) - 2 = 0$

$3x + 1.90 - 2x - 2 = 0$

$\mathrm{x} = 0.1$

$\%\mathrm{Fe^{3+}} = \frac{0.1}{0.95} \times 100$

$= 10.5$

Q.44 Answer:

pressure

Solution:

The maximum quantity of solid that can be dissolved in a given volume of solvent does not depend on the pressure. This is because solid and liquid are completely incompressible, and are essentially unaffected by pressure change.

Q.45 Answer:

8 : 1

Solution:

$r_n = n^2 \times \eta$

$\mathrm{r_2} = 4\eta$

$\mathrm{T_n} = \frac{\mathrm{n^3}}{7^2} \times 1.5 \times 10^{-16}\mathrm{sec}$

$\mathrm{T_1} = \frac{1^3}{1^2} \times 1.5 \times 10^{-16}\mathrm{sec}$

$\mathrm{T_2} = \frac{2^3}{1^2} \times 1.5 \times 10^{-16}\mathrm{sec}$

On dividing $\mathrm{T_2}$ by $\mathrm{T_1}$ we get $\frac{\mathrm{T_2}}{\mathrm{T_1}} = \frac{8}{1}$ or $\frac{\mathrm{T_1}}{\mathrm{T_2}} = \frac{1}{8}$

Q.46 Answer:

—

Solution:

For conjugate-acid base pairs, the acid dissociation constant K_a and base dissociation constant K_b are related by the following equations:

$K_a \cdot K_b = K_w$

Where, K_w is ionization constant for water.

Since $K_a = K_b$ .....(Given)

$K_a{}^2 = K_w K_w = 10^{-14} \Rightarrow K_a = \sqrt{K_w} \Rightarrow K_a = \sqrt{10^{-14}} \Rightarrow K_a = 10^{-7}$

For the reaction

$HA \rightleftharpoons H^+ + A^-$

$K_a = \frac{[H^+][A^-]}{[HA]}$

But here, $[H^+] = [A^-]$

Therefore,

$K_a = \frac{[H^+]^2}{[HA]}$

$[HA] = 0.1\ M$ **.....(assuming no appreciable dissociation)**

$\Rightarrow K_a = 10^{-7} = \frac{[H^+]^2}{0.1} \Rightarrow [H^+] = \sqrt{10^{-7} \times 0.1} \Rightarrow [H^+] = \sqrt{10^{-8}} = 10^{-4}$

We know that,

$pH = -\log[H^+] \Rightarrow pH = -\log 10^{-4} \Rightarrow pH = -(-4 \log\ 10) \Rightarrow pH = 4 \times 1\ (\text{As}\ \log\ 10 = 1) \Rightarrow pH = 4$

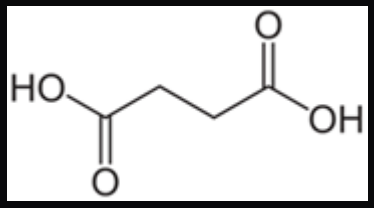
4.0 **is the** pH **of** $0.1\ M$ **solution.**

Q.47 Answer:

—

Solution:

The structure of Succinic Acid doesn't have any chiral carbon.



Hence only one optically optically inactive stereoisomer of succinic acid will be there.

Q.48 Answer:

—

Solution:

$H_3\overset{\overset{O}{||}}{C}-C-H$ **and** H_3C-CH_2-C-H **react with Fehling solution.**

Q.49 Answer:

—

Solution:

Energy released = Energy due to formation of two single bonds $= 2 \times 331 = 662\ \text{kJ mol}^{-1}$ **of propene**

$\Delta \text{H polymerisation /mol} = 590 - 662 = -72\ \text{kJ mol}^{-1}$

$\Delta \text{H polymerisation} = -72 \times \text{n} = -360 \Rightarrow \text{n} = 5$

Q.50 Answer:

—

Solution:

When a non-volatile solute is added in a pure solvent, the freezing point of the solution decreases. This decrease in the freezing point is known as depression in freezing point.

Here,

Weight of solute, $\text{W}_\text{B} = 8\ \text{g}$

Weight of solvent, $\text{W}_\text{A} = 92\ \text{g} = 0.092\,\text{kg}$

Molal freezing point depression constant, $\text{K}_\text{f} = 1.85\ ^\circ\text{C mol}^{-1}$

Depression in freezing point, $\Delta\text{T}_\text{f} = \text{K}_\text{f}\text{m} \Rightarrow \Delta\text{T}_\text{f} = \text{K}_\text{f} \times \frac{\text{W}_\text{B}}{\text{M}_\text{B}} \times \frac{1}{\text{W}_\text{A}} \Rightarrow \text{M}_\text{B} = \frac{1.85 \times 8}{0.925 \times 0.092} = 173.9$

Hence, the molar mass of nicotine = $174\ \text{g mol}^{-1}$**.**

Q.51 Answer:

None of these

Solution:

The number of digits in the locker dialing is 3-digit.

The limits of the three digits are from 000 to 999.

Let the event that the stranger can succeed at a particular trial is $P(A)$**.**

The equation to find that the stranger success in his trials is,

$P(A) = \frac{999}{1000} \times \frac{998}{999} \times \frac{997}{998} \times \ldots \times \frac{1000-k+1}{1000} \times \frac{1}{1000-k+1}$

Then by solving the above expression, we will get

$$P(A) = \frac{1}{1000}$$
$$= 0.001$$

Now we have to find the value of $1000p$ **then, substituting the value of** p **in the term, we will get**

$$1000p = 1000(0.001)$$
$$= 1$$

Therefore, the value of the $1000p$ **is 1.**

Q.52 Answer:

256

Solution:

Given, $x_1.x_2.x_3.x_4 = 770 = 2.5.7.11$

Let positive integer $x_1 = 2^{a_1}.5^{b_1}.7^{c_1}.11^{d_1}$

Let positive integer $x_2 = 2^{a_2}.5^{b_2}.7^{c_2}.11^{d_2}$

Let positive integer $x_3 = 2^{a_3}.5^{b_3}.7^{c_3}.11^{d_3}$

Let positive integer $x_4 = 2^{a_4}.5^{b_4}.7^{c_4}.11^{d_4}$

Now, $x_1x_2x_3x_4 = 2^{a_1+a_2+a_3+a_4}.5^{b_1+b_2+b_3+b_4}.7^{c_1+c_2+c_3+c_4}.11^{d_1+d_2+d_3+d_4}$

As per given condition $a_1 + a_2 + a_3 + a_4 = 1$, which can be in 4 ways $(1, 0, 0, 0)$, $(0, 1, 0, 0)$ $(0, 0, 1, 0)$ and $(0, 0, 0, 1)$, similarly for other powers.

Hence total number of ways $= 4 \times 4 \times 4 \times 4 = 256$

Q.53 Answer:

$5\sqrt{3}(\sqrt{6} - 5)$

Solution:

$$\begin{vmatrix} \sqrt{13}+\sqrt{3} & 2\sqrt{5} & \sqrt{5} \\ \sqrt{15}+\sqrt{26} & 5 & \sqrt{10} \\ 3+\sqrt{65} & \sqrt{15} & 5 \end{vmatrix}$$

Take $\sqrt{5}$ common from each of C_2 and C_3 and rearrange the elements of first column

$$\Delta = \sqrt{5} \cdot \sqrt{5} \begin{vmatrix} \sqrt{13}+\sqrt{3} & 2 & 1 \\ \sqrt{26}+\sqrt{15} & \sqrt{5} & \sqrt{2} \\ \sqrt{65}+3 & \sqrt{3} & \sqrt{5} \end{vmatrix}$$

Split into two determinants the first of which will Vanish. Take $\sqrt{13}$ common from Δ_1 and $\sqrt{3}$ from Δ_2 .

$$\therefore \Delta = 0 + 5\sqrt{3} \begin{vmatrix} 1 & 2 & 1 \\ \sqrt{5} & \sqrt{5} & \sqrt{2} \\ \sqrt{3} & \sqrt{3} & \sqrt{5} \end{vmatrix}$$

Apply $\mathrm{C}_1 \to \mathrm{C}_1 - \mathrm{C}_2$ we get

$$\Delta = 5\sqrt{3} \begin{vmatrix} -1 & 2 & 1 \\ 0 & \sqrt{5} & \sqrt{2} \\ 0 & \sqrt{3} & \sqrt{5} \end{vmatrix} = -5\sqrt{3}(5 - \sqrt{6}) = 5\sqrt{3}(\sqrt{6} - 5).$$

Q.54 Answer:

$\frac{2}{11}$

Solution:

$t = 2^{11x} \Rightarrow \frac{t^3}{4} + 4t = 2t^2 + 1, t^3 - 8t^2 + 16t - 4 = 0$ has roots t_1, t_2, t_3

$t_1t_2t_3 = 4 \Rightarrow 2^{11x_1} \cdot 2^{11x_2} \cdot 2^{11x_3} = 4 = 2^2$

$11\left(x_1 + x_2 + x_3\right) = 2 \therefore x_1 + x_2 + x_3 = \frac{2}{11}$

Q.55 Answer:

$\frac{8}{31}$

Solution:

$2^{78} = 2^3 \cdot 2^{75} = 8\left(2^5\right)^{15} = 8(31 + 1)^{15}$

$= 8\left({}^{15}C_0 31^{15} + {}^{15}C_1 31^{14} + \ldots\ldots\ldots + {}^{15}C_{14} 31 + {}^{15}C_{15}\right)$

$2^{78} = 8(31k + 1)$

$\frac{2^{78}}{31} = 8k + \frac{8}{31}$

Q.56 Answer:

$y^2 - x^2 - 2xy + 1 = 0$

Solution:

Let, $\angle RPQ = \theta$ & $\angle RQP = \phi$, where $\theta - \phi = 45^\circ$

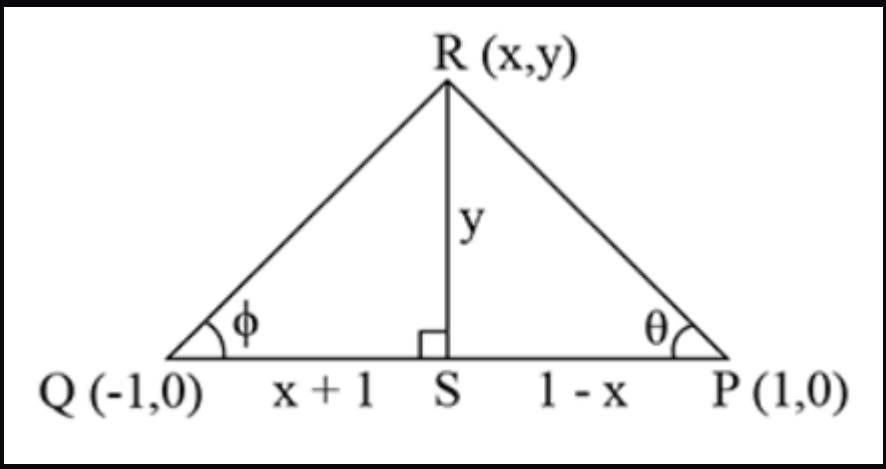
Now, $\tan(\theta - \phi) = 1$

$\Rightarrow \frac{\tan\theta - \tan\phi}{1 + \tan\theta\tan\phi} = 1$, where $\tan\theta = \frac{y}{1-x}$ & $\tan\phi = \frac{y}{x+1}$

$\Rightarrow \frac{y}{1-x} - \frac{y}{x+1} = 1 + \left(\frac{y}{1-x}\right)\left(\frac{y}{x+1}\right)$

$\Rightarrow 2yx = 1 - x^2 + y^2$

$\Rightarrow y^2 - x^2 - 2xy + 1 = 0$



Q.57 Answer:

$f(x) = y(x + c)$

Solution:

We have $\frac{dy}{dx} = \frac{f'(x)}{f(x)}y - \frac{y^2}{f(x)}$

$\Rightarrow \frac{dy}{dx} - \frac{f'(x)}{f(x)}y = -\frac{y^2}{f(x)}$

Divide by y^2

$y^{-2}\frac{dy}{dx} - y^{-1}\frac{f'(x)}{f(x)} = -\frac{1}{f(x)}$

Put $y^{-1} = z \Rightarrow -y^{-2}\frac{dy}{dx} = \frac{dz}{dx}$

$-\frac{dz}{dx} - \frac{f'(x)}{f(x)}(z) = -\frac{1}{f(x)} \Rightarrow \frac{dz}{dx} + \frac{f'(x)}{f(x)}z = \frac{1}{f(x)}$

I.F. $= e^{\int \frac{f'(x)}{f(x)}dx} = e^{\log f(x)} = f(x)$

$\therefore$ The solution is $z(f(x)) = \int \frac{1}{f(x)}(f(x))dx + c$

$\Rightarrow y^{-1}(f(x)) = x + c \Rightarrow f(x) = y(x + c)$

Q.58 Answer:

${}^{13}C_3 \times \frac{12!}{5!3!2!}$

Solution:

All AAAAA, CCC, D, EE, F can be arranged in $\frac{12!}{5!3!2!}$ ways. Between the gaps B can be arranged in $^{13}C_3$ ways Hence, the total number of ways $= {}^{13}C_3 \times \frac{12!}{5!3!2!}$

Q.59 Answer:

-3

Solution:

$f(x) = x^3 + x^2 - 5x - 1$

$f'(x) = 3x^2 + 2x - 5 = (x-1)(3x+5)$

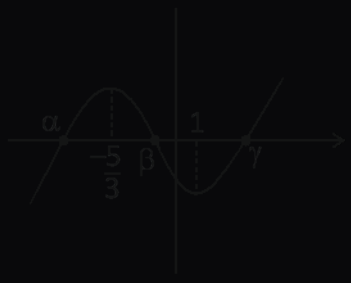


$x = -\frac{5}{3} \rightarrow$ **point of local maxima,**

$x = 1 \rightarrow$ **point of local minima,**

Increasing function in $\left(-\infty, -\frac{5}{3}\right) \cup (1, \infty)$

Decreasing function in $\left(-\frac{5}{3}, 1\right)$



Also, $f(0) = -1, f(1) = -4, f(2) = 1, f(-1) = 5$

$f(-2) = -8 + 4 + 10 - 1 = 5$

$f(-3) = -27 + 9 + 15 - 1 = -4$

$\Rightarrow \alpha \in (-3, -2);\ \beta \in (-1,0);\ \gamma \in (1,2)$

$\Rightarrow [\alpha] = -3,\ [\beta] = -1,\ [\gamma] = 1$

$\Rightarrow [\alpha] + [\beta] + [\gamma] = -3$

Q.60 Answer:

the direction cosines of three mutually perpendicular lines

Solution:

As $(l_1, m_1, n_1), (l_2, m_2, n_2)$ **and** (l_3, m_3, n_3)

the direction cosines of three mutually perpendicular lines

Let $\vec{a} = l_1\hat{i} + m_1\hat{j} + n_1\hat{k}$,

$\vec{b} = l_2\hat{i} + m_2\hat{j} + n_2\hat{k}$ **and** $\vec{c} = l_3\hat{i} + m_3\hat{j} + n_3\hat{k}$

From cofactors of first row

$p_1\hat{i} + q_1\hat{j} + r_1\hat{k} = (m_2n_3 - m_3n_2)\vec{i} + (l_3n_2 - l_2n_3)\vec{j} + (m_3l_2 - m_2l_3)\vec{k}$

Given that $\vec{a}, \vec{b}, \vec{c}$ **are three mutually perpendicular unit vectors.**

Then, $p_1\hat{i} + q_1\hat{j} + r_1\hat{k} = \vec{b} \times \vec{c} = \vec{a}$

$\because$ $\overrightarrow{b} \times \overrightarrow{c}$ parallel to $\overrightarrow{a}$ and $\overrightarrow{b} \times \overrightarrow{c}$ & $\overrightarrow{a}$ are unit vectors.

Similarly, $p_2\hat{i} + q_2\hat{j} + r_2\widehat{k} = \overrightarrow{c} \times \overrightarrow{a} = \overrightarrow{b}$

and $p_3\hat{i} + q_3\hat{j} + r_3\widehat{k} = \overrightarrow{a} \times \overrightarrow{b} = \overrightarrow{c}$

These vectors also mutually perpendicular unit vectors.

Q.61 Answer:

0

Solution:

$\int \frac{1-2019\cos^2 x}{(\sin x)^{2019}\cos^2 x}dx$

$= \int \frac{\sec^2 x dx}{(\sin x)^{2019}} - 2019\int \frac{1}{(\sin x)^{2019}}dx$

Using integration by parts

$= \frac{\tan x}{(\sin x)^{2019}} + C$

$\Rightarrow f\left(\mathrm{x}\right) = \frac{1}{(\mathrm{sinx})^{2019}}$

Q.62 Answer:

$\{0, 1, 3\}$

Solution:

$\mathrm{U} = \left\{x : x^5 - 6x^4 + 11x^3 - 6x^2 = 0\right\}$

$= \{0, 1, 2, 3\}$

$\mathrm{A} = \left\{x : x^2 - 5x + 6 = 0\right\}$

$= \{2, 3\}$

and $B = \left\{x : x^2 - 3x + 2 = 0\right\}$

$= \{2, 1\}$

$\mathrm{A} \cap \mathrm{B} = \{2\}$

$(\mathrm{A} \cap \mathrm{B})' = \mathrm{U} - (\mathrm{A} \cap \mathrm{B})$

$= \{0, 1, 2, 3\} - \{2\} = \{0, 1, 3\}$

Q.63 Answer:

0

Solution:

Given that

$f\left(2x^2 - 1\right) = 2x^3 f(x) \ \ldots (\mathrm{i})$

Replacing x by $-x$ in (i), we get

$f\left(2(-x)^2 - 1\right) = 2(-x)^3 f(-x)$

$\Rightarrow f\left(2x^2 - 1\right) = -2x^3 f(-x) \ \ldots (\mathrm{ii})$

From equations (i) and (ii), we get

$f(x) = \frac{f(2x^2-1)}{2x^3}$ and $f(-x) = \frac{f(2x^2-1)}{-2x^3}$

So, $f(-x) = -f(x)$

i.e., $f(x)$ is an odd function.

So, $f(0) = 0$.

As even order derivative of odd function will also be odd function.

$f^{(2010)}(0) = 0$

Q.64 Answer:

$\left(1 - \frac{1}{e}\right)$

Solution:

$$f(x) = 1 + \frac{\int_1^x f(t)dt}{x}$$

$$xf(x) = x + \int_1^x f(t)dt$$

$$f(x) + xf'(x) = 1 + f(x)$$

$$f'(x) = \frac{1}{x}$$

$$f(x) = \ln_e |x| + C$$

$$\because f(1) = 1 \Rightarrow f(x) = \ln_e |x| + 1$$

Required area

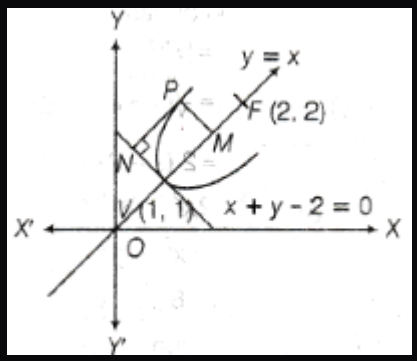
$= \int_0^1 e^{y-1} dy = \left(1 - \frac{1}{e}\right)$

Q.65 Answer:

$(x - y)^2 = 8(x + y - 2)$

Solution:

Since, distance of vertex from origin is $\sqrt{2}$ and focus is $2\sqrt{2}$, here $a = \sqrt{2}$.



$\therefore V(1, 1)$ and $F(2, 2) i.e.$, axis lies on $y = x$.

Length of latusrectum $= 4a = 4\sqrt{2}$

By definition of parabola, $PM^2 = (4a)(PN)$

where, PN is length of perpendicular upon line NV.

$$x + y - 2 = 0$$

$$\Rightarrow \quad \frac{(x - y)^2}{2} = 4\sqrt{2}\left(\frac{x + y - 2}{\sqrt{2}}\right)$$

$$\therefore \quad (x - y)^2 = 8(x + y - 2)$$

Q.66 Answer:

$(\vec{a} \cdot \hat{i})^2 + (\vec{a} \cdot \hat{j})^2 + (\vec{a} \cdot \hat{k})^2 = a^2.$

Solution:

(a) $|\vec{a}| = |b| = |\vec{a} - b| = 1$

$a^2 + b^2 - 2\vec{a} \cdot \vec{b} = 1$

$\therefore$ Angle between $\vec{a}$ & $\vec{b}$ is $\frac{\pi}{3}$

(b) A vector in the direction of angle bisector of vectors $\vec{a}$ and $\vec{b}$ is $\vec{a} + \vec{b}$

$\therefore$ The given statement is not correct.

(c) $(\vec{a} \cdot \hat{i})^2 + (\vec{a} \cdot \hat{j})^2 + (\vec{a} \cdot \hat{k})^2 = |\vec{a}|^2 = a^2$

(d) Any vector in the plane $\hat{i} + \hat{j} + \hat{k}$ and $-\hat{i} + \hat{j} + \hat{k}$ is of the form

$\alpha(\hat{i} + \hat{j} + \hat{k}) + \beta(-\hat{i} + \hat{j} + \hat{k})$

$= (\alpha - \beta)\hat{i} + (\alpha + \beta)\hat{j} + (\alpha + \beta)\hat{k}$

This will be perpendicular with $-\hat{i} + \hat{j} + \hat{k}$

if $-(\alpha - \beta) + (\alpha + \beta) + (\alpha + \beta) = 0 \Rightarrow \alpha = -3\beta$

Hence the required vector is of the form

$(-4\hat{i} - 2\hat{j} - 2\hat{k})$

$\therefore$ statement is false.

Q.67 Answer:

$P = 5, Q = -6, R = 3$

Solution:

$f(x) = Pe^{2x} + Qe^x + Rx$

$\Rightarrow f'(x) = 2Pe^{2x} + Qe^x + R$

$\Rightarrow 31 = 2\mathrm{P}e^{2\ln 2} + \mathrm{Q}e^{\ln 2} + \mathrm{R}$

$\Rightarrow 8\mathrm{P} + 2\mathrm{Q} + \mathrm{R} = 31 \ldots$ (i)

Also, $-1 = \mathrm{P} + \mathrm{Q} \ldots$ (ii)

& $\int_0^{\log 4} (f(x) - Rx)dx = \frac{39}{2}$

$\Rightarrow \int_0^{\log 4} \left(\mathrm{P}e^{2x} + \mathrm{Q}e^{x}\right) \mathrm{d}x = \frac{39}{2}$

$\Rightarrow \left[\frac{\mathrm{P}}{2}e^{2x} + \mathrm{Q}e^{x}\right]_0^{\log 4} = \frac{39}{2}$

$\Rightarrow \frac{P}{2}e^{2\log 4} + Qe^{\log 4} - \frac{P}{2} - Q = \frac{39}{2}$

(iii) $\Rightarrow 15\mathrm{P} + 6\mathrm{Q} = 39$

Solving (i), (ii) and (iii), we get $\mathrm{P} = 5, \mathrm{Q} = -6, \mathrm{R} = 3$

Q.68 Answer:

$\tan^{-1}(2n)$

Solution:

$y = \tan^{-1}\frac{1}{1+x+x^2} + \tan^{-1}\frac{1}{x^2+3x+3} + \ldots + 2n$ terms

$= \tan^{-1}\frac{(x+1)-x}{1+x(1+x)} + \tan^{-1}\frac{(x+2)-(x+1)}{1+(x+1)(x+2)} + \ldots\ldots$ (2n terms)

$= \tan^{-1}(x+1) - \tan^{-1}x + \tan^{-1}(x+2) - \tan^{-1}(x+1) + \ldots + \tan^{-1}(x+2n) - \tan^{-1}(x+(2n-1))$

$= \tan^{-1}(x+2n) - \tan^{-1} x$
$y(0) = \tan^{-1}(2n)$

Q.69 Answer:

$A \geq -\frac{1}{2}$

Solution:

On squaring both sides

$(a_i)^2 = (a_{i-1}+1)^2$

$a_i^2 = (a_{i-1})^2 + 1 + 2a_{i-1}$

$a_i^2 - a_{i-1}^2 = 2a_{i-1} + 1$

Substituting $i = 2, 3, \ldots, n+1$

$$\begin{array}{l} a_2^2 - a_1^2 = 2a_1 + 1 \\ a_3^2 - a_2^2 = 2a_2 + 1 \\ a_4^2 - a_3^3 = 2a_3 + 1 \\ \quad | \\ \quad\quad | \\ a_{n+1}^2 - a_n^2 = 2a_n + 1 \\ \hline a_{n+1}^2 - a_1^2 = 2\left(a_1 + a_2 + a_3 + \ldots\ldots + a_n\right) + n \end{array}$$

Let A.M. of $(a_1, a_2, a_3, \ldots\ldots, a_n) = A$

$A = \dfrac{a_1 + a_2 + a_3 + \ldots + a_n}{n}$

$nA = a_1 + a_2 + a_3 + \ldots + a_n$

$a_{n+1}^2 - a_1^2 = 2\left(a_1 + a_2 + a_3 + \ldots\ldots + a_n\right) + n$

Given $a_1 = 0$

So, $a_{n+1}^2 - 0 = 2(nA) + n$

$\dfrac{a_{n+1}^2 - n}{2n} = A$

$A = \dfrac{a_{n+1}^2}{2n} - \dfrac{1}{2}$

$A \geq -\dfrac{1}{2}\left(\because \dfrac{a_{n+1}^2}{2n} \text{ is always positive}\right)$

Q.70 Answer:

$\frac{24}{\sqrt{34}}$

Solution:

$f(g(x)) = x \Rightarrow f'\left(g(x) \cdot g'(x)\right) = 1 \Rightarrow g'(x) = \frac{1}{f'(g(x))}$

Let $g(6) = a \Rightarrow f(a) = 6 \Rightarrow a^2 - 2a + 3 = 36 \Rightarrow (a-1)^2 = 34$

$\Rightarrow g'(6) = \frac{1}{f'(g(6))} = \frac{1}{f'(a)} = \frac{\sqrt{a^2-2a+3}}{(a-1)} = \frac{6}{\sqrt{34}}$

Now,

$h(x) = g\left(f^2(x)\right) \Rightarrow h'(x) = g'\left(f^2(x)\right) \cdot 2f(x)f'(x) \Rightarrow h'(3) = g'(6) \cdot 2f(3) \cdot f'(3)$
$\therefore h'(3) = 2 \times \frac{6}{\sqrt{34}} \times 2 = \frac{24}{\sqrt{34}}$

Q.71 Answer:

—

Solution:

At $x \to 0^+, \{.\} = x$

$\lim\limits_{x \to 0^+} \frac{x^x \cos\left(x^{\sin x+1}\right)}{\cos(\tan x - x)\cos(x-x)}$

$k = 1,\ as\ \lim\limits_{x \to 0^+} x^x = 1$

Q.72 Answer:

—

Solution:

Given two lines with parametric equations $x = 1 + t,\ y = 1 + 6t,\ z = 2t$ **and**
$x = 1 + 2s,\ y = 5 + 15s,\ z = -2 + 6s$.
So their vector equations are $\vec{r} = \left(\hat{i} + \hat{j} + 0\hat{k}\right) + t\left(\hat{i} + 6\hat{j} + 2\hat{k}\right),\ \vec{r} = \left(\hat{i} + 5\hat{j} - 2\hat{k}\right) + t\left(2\hat{i} + 15\hat{j} + 6\hat{k}\right)$.
We know, The shortest distance between two skew-lines $\vec{r} = \vec{a_1} + \lambda\vec{b_1}$ **and** $\vec{r} = \vec{a_2} + \mu\vec{b_2}$ **is equal to**
$\left|\frac{\left(\vec{a_1} - \vec{a_2}\right) \cdot \left(\vec{b_1} \times \vec{b_2}\right)}{\left|\vec{b_1} \times \vec{b_2}\right|}\right|$
On comparing, we get
$\vec{a_1} = \hat{i} + \hat{j} + 0\hat{k},\ \vec{b_1} = \hat{i} + 6\hat{j} + 2\hat{k},\ \vec{a_2} = \hat{i} + 5\hat{j} - 2\hat{k},\ \vec{b_2} = 2\hat{i} + 15\hat{j} + \hat{k}$.
$\therefore\ \vec{a_1} - \vec{a_2} = \left(\hat{i} + \hat{j} + 0\hat{k}\right) - \left(\hat{i} + 5\hat{j} - 2\hat{k}\right)$
$\Rightarrow\ \vec{a_1} - \vec{a_2} = 0\hat{i} - 4\hat{j} + 2\hat{k}$
And $\vec{b_1} \times \vec{b_1} = \left(\hat{i} + 6\hat{j} + 2\hat{k}\right) \times \left(2\hat{i} + 15\hat{j} + 6\hat{k}\right) \quad \Rightarrow \vec{b_1} \times \vec{b_1} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 6 & 2 \\ 2 & 15 & 6 \end{vmatrix}$
$\Rightarrow\ \vec{b_1} \times \vec{b_1} = 6\hat{i} - 2\hat{j} + 3\hat{k}$
$\therefore$ **The shortest distance between the given lines** $= \left|\frac{\left(\vec{a_1} - \vec{a_2}\right) \cdot \left(\vec{b_1} \times \vec{b_2}\right)}{\left|\vec{b_1} \times \vec{b_2}\right|}\right|$
$\Rightarrow\ \left|\frac{\left(0\hat{i} - 4\hat{j} + 2\hat{k}\right) \cdot \left(6\hat{i} - 2\hat{j} + 3\hat{k}\right)}{\left|6\hat{i} - 2\hat{j} + 3\hat{k}\right|}\right| = 2$

Q.73 Answer:

—

Solution:

Changing into polar, we get

$\alpha_k = e^{\left(ik\frac{\pi}{7}\right)}$

$\alpha_{k+1} - \alpha_k = e^{\left(i(k+1)\frac{\pi}{7}\right)} - e^{ik\frac{\pi}{7}} = e^{ik\frac{\pi}{7}}\left(e^{i\frac{\pi}{7}} - 1\right)$

Also we know that $\left|e^{ik\frac{\pi}{7}}\right| = 1$. So, taking absolute value, we get

$\left|e^{i\frac{\pi}{7}} - 1\right| = \sqrt{\sin^2\left(\frac{\pi}{7}\right) + \left(\cos\left(\frac{\pi}{7}\right) - 1\right)^2}$

The absolute value of difference of two consecutive terms are equal.

$$\frac{\sum_{K=1}^{12}\left|e^{\frac{i(K+1)\pi}{7}} - e^{i\frac{K\pi}{7}}\right|}{\sum_{K=1}^{3}\left|e^{\frac{i(4K-1)\pi}{7}} - e^{i\frac{(4K-2)\pi}{7}}\right|} = \frac{\sum_{K=1}^{12}\left|e^{i\frac{K\pi}{7}}\right|\cdot\left|e^{i\frac{\pi}{7}} - 1\right|}{\sum_{K=1}^{3}\left|e^{\frac{i(4K-1)\pi}{7}}\right|\cdot\left|e^{\frac{i\pi}{7}} - 1\right|} = \frac{\sum_{K=1}^{12}\left|e^{\frac{i\pi}{7}} - 1\right|}{\sum_{K=1}^{3}\left|e^{\frac{i\pi}{7}} - 1\right|} = \frac{12\left|e^{\frac{i\pi}{7}} - 1\right|}{3\left|e^{\frac{i\pi}{7}} - 1\right|} = 4$$

Q.74 Answer:

—

Solution:

Mean $(X_1, X_2 \ldots X_n) = \frac{\frac{n}{2}[2X_1 + (n-1)d]}{n} = 5$

Mean of $(y_1, y_2 \ldots . y_n) = \frac{\frac{y_1(16^n - 1)}{15}}{n} = 5$

$2X_1 + (n-1)d = 10 \qquad \ldots(1)$

$y_1(16^n - 1) = 75n \qquad \ldots(2)$

Mean of $(X_6, X_7 \ldots . X_{n+5})$ = Mean of $(y_{\mathrm{P}+1}, y_{\mathrm{P}+2}, y_{\mathrm{P}+n})$

$\frac{\frac{n}{2}[2X_6 + (n-1)d]}{n} = \frac{y_{P+1}(16^n - 1)}{15n} = \frac{y_1 16^P(16^n - 1)}{15n}$

$X_6 + \left(\frac{n-1}{2}\right)d = \frac{16^P(75n)}{15n} = 5 \times 16^P$

$X_1 + 5d + \left(\frac{n-1}{2}\right)d = 5 \times 16^P$

$5 - \left(\frac{n-1}{2}\right)d + 5d + \left(\frac{n-1}{2}\right)d = 5 \times 16^P \Rightarrow d = 16^P - 1$

$\because d$ is 2 digit natural number

$\Rightarrow P = 1,\ d = 15$

Q.75 Answer:

—

Solution:

$4\cos 36^\circ = 4\left(\frac{\sqrt{5}+1}{4}\right) = \sqrt{5} + \sqrt{1}$

$\cot 7\frac{1}{2}^\circ = \frac{1 + \cos 15^\circ}{\sin 15^\circ} = \frac{1 + \left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right)}{\left(\frac{\sqrt{3}-1}{2\sqrt{2}}\right)} = \frac{2\sqrt{2} + \sqrt{3} + 1}{\sqrt{3} - 1}$

$= \frac{(2\sqrt{2} + \sqrt{3} + 1)(\sqrt{3} + 1)}{(\sqrt{3} - 1)(\sqrt{3} + 1)} = \frac{2\sqrt{6} + 2\sqrt{2} + 3 + \sqrt{3} + \sqrt{3} + 1}{2}$

$= \frac{2\sqrt{6} + 2\sqrt{3} + 2\sqrt{2} + 4}{2} = \sqrt{6} + \sqrt{3} + \sqrt{2} + 2 = \sqrt{6} + \sqrt{4} + \sqrt{3} + \sqrt{2}$

$\Rightarrow 4\cos 36^\circ + \cot\left(7\frac{1}{2}^\circ\right) = \sqrt{5} + \sqrt{1} + \sqrt{6} + \sqrt{4} + \sqrt{3} + \sqrt{2}$

$= \sqrt{1} + \sqrt{2} + \sqrt{3} + \sqrt{4} + \sqrt{5} + \sqrt{6}$

$= \sqrt{n_1} + \sqrt{n_2} + \sqrt{n_3} + \sqrt{n_4} + \sqrt{n_5} + \sqrt{n_6}$

$\Rightarrow \sum_{i=1}^{6} n_i^2 = 1^2 + 2^2 + 3^3 + 4^2 + 5^2 + 6^2$

$= \frac{6 \times 7 \times 13}{6} = 91$

$\Rightarrow \frac{\sum_{i=1}^{6} n_i^2}{7} = \frac{91}{7} = 13$